Supplementary Information

Survival-supervised deconvolution identifies a replicable tumor–stroma prognostic architecture in pancreatic cancer

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1 Model details

Let $X \in \mathbb{R}_{\geq 0}^{p\times n}$ denote the nonnegative gene expression matrix with p genes and n subjects. DeSurv approximates $X \approx WH$ with $W \in \mathbb{R}_{\geq 0}^{p\times k}$ and $H \in \mathbb{R}_{\geq 0}^{k\times n}$, and links the shared program matrix W to survival via an elastic-net–penalized Cox model. Supervision is placed on the program matrix W, the basis that transfers to new cohorts, whereas the sample loadings H serve only to reconstruct the training expression matrix; new samples are therefore scored by projection onto the learned programs, $Z = W^\top X$, rather than by re-estimating loadings.

Let $y \in \mathbb{R}_{>0}^{n}$ be the observed survival times, $\delta \in \{0,1\}^n$ the event indicators, and $Z = W^\top X \in \mathbb{R}^{k\times n}$ the sample-wise factor scores. We write Z_i for the ith column of Z, and $n_{\text{event}} = \sum_{i=1}^{n} \delta_i$. For convenience we recall the DeSurv loss from Equation 1 in the main text and rewrite it as

$$\begin{aligned}\mathcal{L}(W,H,\beta) = \\ &(1-\alpha)\left(\frac{1}{2\|X\|_F^2}\|X - WH\|_F^2 + \frac{\lambda_H}{2nk}\|H\|_F^2\right) \\ &-\alpha\left(\frac{2}{n_{event}}\ell(W,\beta) - \lambda\{\xi\|\beta\|_1 + \tfrac{1-\xi}{2}\|\beta\|_2^2\}\right)\end{aligned}$$

where $\|\cdot\|_F$ is the Frobenius norm and $\beta \in \mathbb{R}^k$ are the Cox coefficients. The Cox log-partial likelihood $\ell(W,\beta)$ is

$$\ell(W,\beta) = \sum_{i=1}^{n} \delta_i \left[Z_i^\top \beta - \log\left(\sum_{j:y_j \geq y_i} \exp(Z_j^\top \beta) \right) \right]. \tag{1}$$

The inner sum runs over the risk set $\{j : y_j \geq y_i\}$ of subjects still at risk at the observed time y_i; tied event times are handled by the Breslow approximation in the `glmnet` Coxnet routine used for the β update (below). Hyperparameters satisfy $\alpha \in [0,1)$, $\lambda_H \geq 0$, $\lambda \geq 0$, and $\xi \in [0,1]$. The constants $1/(2\|X\|_F^2)$, $2/n_{\text{event}}$, and $1/(2nk)$ place the reconstruction, loadings-penalty, and survival terms on comparable numerical scales for stable optimization.

2 Optimization Algorithm

2.1 Block coordinate descent scheme

Optimization proceeds by block coordinate descent (BCD; Algorithm S1), which at each outer iteration updates H, then W, then β while holding the other blocks fixed.

2.2 Update for H

Conditional on W, the loss $\mathcal{L}(W,H,\beta)$ reduces to a convex quadratic function of H. We adopt the standard NMF multiplicative update with ℓ_2 penalty:

$$H \leftarrow \max\left(H \odot \frac{W^\top X}{W^\top WH + \frac{\|X\|_F^2}{nk}\lambda_H H + \varepsilon_H}, \varepsilon_H \right), \tag{2}$$

Algorithm S1 DeSurv block coordinate descent

Input: $X \in \mathbb{R}^{p\times n}_{\geq 0}$, survival times $y \in \mathbb{R}^{n}_{>0}$, event indicators $\delta \in \{0,1\}^n$, tolerance tol, maximum iterations $maxit$, supervision parameter α, rank k, penalties λ_H, λ, and ξ
Output: Fitted DeSurv parameters (W, H, β)

Initialize $W^{(0)}, H^{(0)}$ with positive entries (e.g., $W^{(0)}, H^{(0)} \sim \text{Uniform}(0, \max X)$)
Initialize $\beta^{(0)}$ (e.g., $\beta^{(0)} = \mathbf{1}$)
$loss = \mathcal{L}(W^{(0)}, H^{(0)}, \beta^{(0)})$
$eps = \infty$, $t = 0$
while $eps \geq tol$ **and** $t < maxit$ **do**
  (H-update)
    Update $H^{(t+1)}$ from $H^{(t)}$ using the multiplicative rule in Eq. 2,
    holding $W^{(t)}$ and $\beta^{(t)}$ fixed.
  (W-update)
    Update $W^{(t+1)}$ from $W^{(t)}$ using the hybrid multiplicative rule in Eq. 6,
    holding $H^{(t+1)}$ and $\beta^{(t)}$ fixed, and perform backtracking to ensure
    $\mathcal{L}$ does not increase.
  (β-update)
    Update $\beta^{(t+1)}$ by solving the elastic-net Cox subproblem using cyclic coordinate descent to numerical tolerance
    (Eq. 7), holding $W^{(t+1)}$ and $H^{(t+1)}$ fixed.
  $lossNew = \mathcal{L}(W^{(t+1)}, H^{(t+1)}, \beta^{(t+1)})$
  $eps = |lossNew - loss|/|loss|$
  $loss = lossNew$
  $t = t + 1$
end while
return $\hat{W} = W^{(t)}, \hat{H} = H^{(t)}, \hat{\beta} = \beta^{(t)}$

where $\odot$ denotes elementwise multiplication, all divisions are elementwise, and $\varepsilon_H > 0$ is a small floor to prevent entries from becoming exactly zero. This update is equivalent to a majorization–minimization step and guarantees a nonincreasing reconstruction term conditional on W[1,2]. The ε_H-floor ensures that iterates remain in the interior of the nonnegative orthant, which keeps the multiplicative H-update numerically well-defined.

2.3 Update for W

For W, we construct a hybrid multiplicative update that balances the contributions of the NMF and Cox gradients. Let

$$\nabla_W \mathcal{L}_{\text{NMF}}(W, H) = \frac{1}{\|X\|_F^2}(WH - X)H^\top,$$

and let

$$\nabla_W \mathcal{L}_{\text{Cox}}(W, \beta) = \frac{2}{n_{event}} \nabla_W \ell(W, \beta).$$

where $\nabla_W \ell(W, \beta)$ denotes the Cox gradient with respect to W. A closed-form expression for $\nabla_W \ell$ is

$$\nabla_W \ell(W, \beta) = \sum_{i=1}^{n} \delta_i \left(x_i - \frac{\sum_{j:y_j \geq y_i} x_j \exp(x_j^\top W \beta)}{\sum_{j:y_j \geq y_i} \exp(x_j^\top W \beta)} \right) \beta^\top, \tag{3}$$

where x_i denotes the ith column of X.

We define a gradient-scaling heuristic

$$\zeta^{(t)} = \frac{\left\|(1-\alpha)\nabla_W \mathcal{L}_{\text{NMF}}(W^{(t)}, H^{(t+1)})\right\|_F^2}{\left\|\alpha \nabla_W \mathcal{L}_{Cox}(W^{(t)}, \beta^{(t)})\right\|_F^2}, \tag{4}$$

and clip $\zeta^{(t)}$ to avoid extreme values: $\zeta^{(t)} \leftarrow \min(\zeta^{(t)}, \zeta_{\max})$ with $\zeta_{\max} = 10^6$ in all experiments.

The multiplicative factor for W is then

$$R^{(t)} = \frac{\frac{(1-\alpha)}{\|X\|_F^2} X H^{(t+1)\top} + \frac{2\alpha}{n_{\text{event}}} \zeta^{(t)} \nabla_W \ell(W^{(t)}, \beta^{(t)})}{\frac{(1-\alpha)}{\|X\|_F^2} W^{(t)} H^{(t+1)} H^{(t+1)\top} + \varepsilon_W}, \tag{5}$$

and the proposed update is

$$W^{(t+1)} = \max\Big(W^{(t)} \odot R^{(t)}, \varepsilon_W\Big), \tag{6}$$

with a small floor $\varepsilon_W > 0$. When $\alpha = 0$, this reduces to the standard multiplicative update for NMF[1]; for $\alpha > 0$, the Cox gradient perturbs the update in a direction that decreases the supervised loss.

Because the combined multiplicative step does not, by itself, guarantee a nonincrease in the full loss $\mathcal{L}$, we embed it in a backtracking line search.

Algorithm S2 W update with backtracking

Input: Value of W and the previous iteration t: $W^{(t)}$, the multiplicative update term at the current iteration $R^{(t)}$, the max number of backtracking updates max_bt, the backtracking parameter $\rho \in (0, 1)$

Output: $W^{(t+1)}$

$\theta = 1$
b = 1
$flag\_accept = FALSE$
while $b \leq max\_bt$ **do**
    $W_{cand}^{(t+1)} = \max\big(W^{(t)} \odot [(1-\theta) + \theta R^{(t)}],\ \varepsilon_W\big)$
    (Column normalization)
        Compute $D = \text{diag}\big(\|W_{(:,1)cand}^{(t+1)}\|_2, \dots, \|W_{(:,k)cand}^{(t+1)}\|_2\big)$
        and set $W_{cand}^{(t+1)} \leftarrow W_{cand}^{(t+1)} D^{-1}$, $H_{cand}^{(t+1)} \leftarrow D H^{(t+1)}$, and $\beta_{cand}^{(t)} \leftarrow D\beta^{(t)}$
    if $\mathcal{L}(W_{cand}^{(t+1)}, H^{(t+1)}, \beta^{(t)}) \leq \mathcal{L}(W^{(t)}, H^{(t+1)}, \beta^{(t)})$ **then**
        $W^{(t+1)} = W_{cand}^{(t+1)}$
        $H^{(t+1)} = H_{cand}^{(t+1)}$
        $\beta^{(t)} = \beta_{cand}^{(t)}$
        $flag\_accept = TRUE$
        break
    end if
    $\theta = \theta * \rho$
    $b = b + 1$
end while
if $flag\_accept = FALSE$ **then**
    $W^{(t+1)} = W^{(t)}$
end if

Column normalization preserves the reconstruction WH and linear predictor $W\beta$. Because the accompanying rescaling can alter penalization terms, the normalized proposal is evaluated using the full penalized objective within the backtracking procedure, which accepts the update only if it does not increase $\mathcal{L}$.

2.4 Update for β

Conditional on (W, H), the loss in β reduces to a convex elastic-net–penalized Cox problem; the common factor $\alpha > 0$ multiplies both the log-partial-likelihood and the penalty in the joint loss (Equation 1) and is dropped here, as it does not affect the minimizer:

$$\min_{\beta \in \mathbb{R}^k} \left\{ -\frac{2}{n_{\text{event}}} \ell(W,\beta) + \lambda \left(\xi \|\beta\|_1 + \frac{1-\xi}{2} \|\beta\|_2^2 \right) \right\}.$$

We solve this subproblem by cyclic coordinate descent following[3]. Writing $\ell(\beta) = \ell(W,\beta)$ and $\tilde{\eta}_i = Z_i^\top \beta$, the update for coordinate r has the closed form

$$\hat{\beta}_r = \frac{S\left(\frac{1}{n} \sum_{i=1}^n w(\tilde{\eta})_i z_{i,r} \left[v(\tilde{\eta})_i - \sum_{j \neq r} z_{i,j} \beta_j \right], \lambda \xi \right)}{\frac{1}{n} \sum_{i=1}^n w(\tilde{\eta})_i z_{i,r}^2 + \lambda(1-\xi)}, \tag{7}$$

where $S(a,\tau) = \operatorname{sign}(a) \max(|a| - \tau, 0)$ is the soft-thresholding operator, and $w(\tilde{\eta})$, $v(\tilde{\eta})$ are standard local quadratic approximations of the Cox log-partial likelihood[3]. We iterate coordinate updates until the subproblem converges to numerical tolerance, yielding a minimizer in β for the current W.

2.5 Normalization of W

After each accepted W update, we normalize the columns of W as $W \leftarrow WD^{-1}$, and adjust H and β as

$$H \leftarrow DH, \qquad \beta \leftarrow D\beta,$$

where D is the diagonal matrix of column ℓ_2-norms of W. This preserves the reconstruction WH and the linear predictor $W\beta$; because the accompanying rescaling of H and β can alter the penalization terms, the normalized proposal is evaluated under the full penalized objective within the backtracking procedure for the W update rather than assumed loss-invariant. Normalization prevents degeneracy in the scale-nonidentifiable factorization and keeps columns of W comparable in magnitude and interpretable as gene programs.

3 Supervision on W versus H

Classical nonnegative matrix factorization (NMF) is non-identifiable: for any invertible matrix R with $WR \geq 0$ and $R^{-1}H \geq 0$, the factorization (W,H) can be transformed to $(WR, R^{-1}H)$ without changing the reconstruction error, yielding multiple valid solutions[4–6]. Although normalizing columns of W or rows of H resolves the trivial scaling ambiguity, nonnegative mixing transformations persist, so factorizations started from different initializations need not agree[7–9]. This applies to DeSurv as well: individual restarts of the reported model are dispersed, which is why the final fit is obtained by a consensus procedure over many restarts (Stability of the recovered gene programs).

In DeSurv, survival supervision enters through the projection $Z = W^\top X$ in the partial log-likelihood, so that the gradient of the DeSurv loss with respect to the programs is

$$\nabla \mathcal{L} = (1-\alpha) \nabla_W \mathcal{L}_{\text{NMF}}(W,H) - \alpha \nabla_W \mathcal{L}_{\text{Cox}}(W,\beta),$$

and therefore the update rule for W as in Equation 6 relies on both the reconstruction term and the supervision term. As such, the gene-program definitions are directly shaped by their association with survival. This mechanism amplifies genes aligned with the hazard gradient and suppresses those that dilute prognostic structure, favoring programs enriched for survival-relevant variation.

By contrast, if supervision were applied to the sample loadings H, the model could reduce the Cox loss by redistributing patient-specific coefficients while leaving the gene-level programs W nearly unchanged, a behavior documented for supervised topic models, where supervision applied only to the coefficient matrix modifies H but not the basis W[10].

Supervising W also provides a practical advantage for **portability**: because W defines gene-level programs, new samples can be embedded by the closed-form projection $Z_{\text{new}} = W^\top X_{\text{new}}$. In contrast, supervising

H would not yield such a mapping; instead, obtaining sample loadings for external datasets would require solving a nonnegative least-squares (NNLS) problem for each new sample, introducing optimization noise and reducing reproducibility.

Together, these considerations motivate supervising through W, which shapes the gene programs toward prognostic directions and yields biologically interpretable signatures that generalize across cohorts.

4 Cross-validation procedure

Algorithm S3 describes the cross validation procedure with F folds and R initializations. We define the c-index using comparable pairs $\mathcal{P} = \{(i,j) : y_i < y_j,\ \delta_i = 1\}$ and linear predictors $\hat{\eta}_i$:

$$\hat{c} = \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \left[\mathbb{1}(\hat{\eta}_i > \hat{\eta}_j) + \tfrac{1}{2}\mathbb{1}(\hat{\eta}_i = \hat{\eta}_j)\right]. \tag{8}$$

Algorithm S3 Cross-validation for DeSurv pipeline

Input:
Number of folds F
Number of initializations R
Observed data (X, y, δ)
Hyperparameters k, α, λ, ξ, λ_H, and number of top genes per factor n_{top}

Output: The cross-validated c-index

```
Divide subjects into F folds.
for f = 1 to F do
    Split data into training and validation (X_(−f), y_(−f), δ_(−f)), and (X_(f), y_(f), δ_(f)) sets
    for r = 1 to R do
        set.seed(r)
        Apply Algorithm S1 with inputs (X_(−f), y_(−f), δ_(−f)), and (k, α, λ, ξ, λ_H)
        Obtain Ŵ and β̂ as output from Algorithm S1
        Restrict Ŵ to the top n_top genes per factor (by factor-specificity score) and compute the estimated
    linear predictor on this support: η̂ = X_(f)^T Ŵ β̂
        Compute c-index, denoted ĉ_(f)r, according to Equation 8
    end for
    end loop over r
    Compute average c-index across initializations: ĉ_(f) = (1/R) Σ_{r=1}^R ĉ_(f)r
end for
end loop over f
Compute final cross-validated c-index: ĉ = (1/F) Σ_{f=1}^F ĉ_(f)
return Final estimate ĉ
```

5 Bayesian Optimization

We treat hyperparameter tuning as a black-box optimization problem over a bounded domain. Let θ collect the tuned parameters (at minimum $\theta = (k, \alpha, \lambda, \xi)$, and optionally discrete choices such as the signature size n_{top}). For a given θ we run the cross-validation procedure in Algorithm S3, averaging the c-index across folds and random initializations to form the objective

$$\hat{c}(\theta) = \frac{1}{F}\sum_{f=1}^{F} \hat{c}_{(f)}(\theta).$$

Because $\hat{c}(\theta)$ is expensive to evaluate and non-differentiable with respect to θ, we use Bayesian optimization (BO) to adaptively select candidate configurations.

We place a Gaussian-process (GP) prior on the response surface $f(\theta) \sim \mathcal{GP}(0, k(\theta, \theta'))$ with a Mat'ern kernel and automatic relevance determination length-scales. Continuous parameters are rescaled to $[0, 1]$ and penalties are optimized on a log scale; integer-valued coordinates (e.g., k or n_{top}) are proposed in the continuous space and rounded inside the evaluator. After an initial batch of evaluations, the GP is refit after each new observation and used to guide acquisition.

At iteration t, let $\mu_t(\theta)$ and $\sigma_t(\theta)$ denote the GP posterior mean and standard deviation, and let $f_\star$ be the best observed score. We select new candidates by maximizing expected improvement (EI),

$$\mathrm{EI}(\theta) = (\mu_t(\theta) - f_\star - \kappa)\, \Phi(Z(\theta)) + \sigma_t(\theta)\phi(Z(\theta)), \quad Z(\theta) = \frac{\mu_t(\theta) - f_\star - \kappa}{\sigma_t(\theta)},$$

where Φ and ϕ are the standard normal cdf/pdf and $\kappa \geq 0$ controls the exploration margin. EI is evaluated over a candidate pool sampled from the bounded search region, and the maximizer is evaluated next. The loop continues until the iteration budget is exhausted or the improvement plateaus.

For the PDAC application, the objective was the mean 5-fold cross-validated Harrell C-index (Algorithm S3), with folds stratified by cohort under a fixed seed (123); each configuration was evaluated from 30 random initializations and its C-index averaged across them. The bounded search domain was $k \in [2, 12]$, $\alpha \in [0, 1]$, $\lambda \in [10^{-3}, 10^3]$ (optimized on a $\log_{10}$ scale), $\xi \in [0, 1]$, and $n_{\text{top}} \in [50, 300]$, with the loadings ridge penalty fixed at $\lambda_H = 0$. Bayesian optimization used an initial design of 50 evaluations followed by 100 GP-guided iterations (150 total evaluations), maximizing expected improvement over a candidate pool of 4,000 points sampled from the bounded domain at each step with an exploration margin $\kappa = 0.01$. The unsupervised comparator fixed $\alpha = 0$ and tuned the remaining coordinates identically.

For model selection, we take the highest observed $\hat{c}(\theta)$ as the optimal configuration. When fold-level standard errors are available, we apply a one-standard-error rule for k, selecting the smallest rank whose mean performance lies within one standard error of the best. The final DeSurv model is refit at the selected parameters using the consensus-based initialization described above, and the BO-chosen n_{top} (when tuned) is used to truncate each program for downstream analysis.

The parameter n_{top} controls which genes enter the survival linear predictor. After factorization, each sample's score for factor j is computed as $\widetilde{W}_j^\top x$, where $\widetilde{W}_j$ retains only the n_{top} genes with the largest factor-specificity scores s_{ij} (Supplementary Information, Section 8); genes outside this set still shape the factorization but do not enter the linear predictor. BO selects n_{top} jointly with k, α, λ, and ξ by optimizing cross-validated C-index. When $\alpha = 0$ (no survival supervision), gene selection and survival outcomes are decoupled.

Supplementary Fig. S1 illustrates the tuned response surface for the two most influential coordinates in the PDAC training cohort: the cross-validated C-index over factorization rank k and supervision strength α. Performance is maximized at intermediate supervision and moderate rank and degrades toward the unsupervised corner ($\alpha = 0$), consistent with a benefit of survival supervision; this is an empirical model-selection sensitivity surface, not a simulation result.

5.1 Consensus initialization for the final model

The final DeSurv model at the selected hyperparameters was initialized from a consensus of multiple independent runs to reduce sensitivity to random starts. We first fit $R = 100$ DeSurv models to the training data from independent random nonnegative initializations at the selected $(k, \alpha, \lambda, \xi, n_{\text{top}})$ (seed 1–100). For each run, the top n_{top} genes of every learned program were extracted by factor-specificity score, and a gene-by-gene consensus matrix was accumulated across runs, recording how often each pair of genes appeared together among the top genes of the same program. Genes appearing in the top sets of fewer than $\lceil 0.3R \rceil$ of the R runs were treated as inactive and excluded from clustering. The consensus matrix over the remaining genes was partitioned into k groups by agglomerative hierarchical clustering (average linkage), and each cluster defined one column of the initial gene-program matrix W_0; inactive genes and otherwise-empty entries were set to a small positive floor ($\sqrt{\epsilon_{\text{mach}}}$) to preserve strict positivity. The initial loadings H_0 were obtained by nonnegative least squares given W_0 and the training expression matrix, and the initial Cox coefficients β_0 were set to zero. This consensus seed used only training-cohort expression and was applied solely to initialize the single final refit after hyperparameter selection; it used no validation-cohort data and therefore introduced no information leakage.

6 PDAC Datasets

We analyzed PDAC cohorts from TCGA-PAAD and CPTAC-PDAC as training data[11,12]. External validation used independent cohorts from Dijk, Moffitt, PACA-AU (array and RNA-seq), and Puleo[13–16]. Training models were fit on the combined TCGA and CPTAC expression matrices after harmonizing gene identifiers; validation datasets were processed with the same gene set and transformations before projection, scoring and downstream validation analyses.

Dataset-specific inclusion and preprocessing steps were as follows. For TCGA-PAAD, we excluded samples lacking tumor grade and mapped features to gene symbols. For CPTAC, we retained samples annotated as PDAC by histology. For Dijk, all 90 samples passed quality-control filters and were retained. For Moffitt, only primary tumor specimens were kept. For PACA-AU array and RNA-seq, we retained primary PDAC tumors and collapsed duplicated gene symbols by median; the RNA-seq cohort received a "_seq' ' suffix on sample IDs to avoid collisions with the array cohort. Puleo required no additional dataset-specific filtering beyond survival QC.

Across cohorts, samples without valid survival outcomes (nonpositive time, missing time, or missing event indicator) were removed. Expression matrices were analyzed on a log scale: RNA-seq cohorts as $\log_2(\text{TPM} + 1)$; microarray cohorts in platform-normalized log-scale intensities. For each training cohort (TCGA-PAAD and CPTAC) separately, genes were ranked by mean expression and by variance independently; the 3,000 genes with the lowest sum of these two ranks were retained, selecting genes that are both highly expressed and highly variable. The intersection of these two gene sets yielded 1,970 genes used for model training. Validation datasets were restricted to this same 1,970-gene set. A within-sample rank transformation was then applied to every dataset to mitigate platform and scale effects. Specifically, for each sample the expression values across the 1,970 retained genes were replaced by their ascending ranks (ties resolved by average ranks), so that every sample carries the same nonnegative rank distribution over the shared gene set; ranks are inherently nonnegative and therefore preserve NMF compatibility. Validation cohorts were rank-transformed by exactly the same per-sample procedure before projection, with no use of validation outcomes.

Table S1 summarizes the cohorts, platforms, sample sizes after quality control, and patient-level metadata used in this study. Overall survival time and event indicators were available for all cohorts. PurIST[17] and DeCAF[18] molecular classifications, applied to bulk expression data as described in those references, were used as covariates in the covariate-adjusted external validation analysis. Gene expression data and molecular classifications for the validation cohorts were originally compiled for the DeCAF study[18]; training data were obtained directly from GDC.

7 Survival Analysis

Survival analyses used overall survival time with event indicators as provided by each cohort. For fitted DeSurv models, sample-level program scores were computed as $Z = \widetilde{W}^\top X$ and standardized using the training mean and standard deviation before constructing the linear predictor $\hat{\eta} = Z\hat{\beta}$. For the per-factor associations shown in the validation forest plot, each individual factor score was standardized to zero mean and unit standard deviation within its own cohort, so that per-factor hazard ratios are expressed per one standard-deviation increment; the multivariable linear predictor was likewise reported per standard deviation. Prognostic performance was summarized using the concordance index as defined above. When multiple datasets were pooled, Cox models were stratified by dataset to allow cohort-specific baseline hazards. Continuous-score associations were evaluated using Cox models stratified by dataset.

8 Gene Program Correlation Analysis

To assess the biological identity of learned factors, we evaluated the correspondence between factor-specific gene rankings and established PDAC gene programs. Each column j of the gene program matrix W was first scaled by its column maximum, $W^*_{ij} = W_{ij} / \max_i W_{ij}$, placing all factors on a common $[0, 1]$ scale, and the specificity score for gene i in factor j was defined as

$$s_{ij} = W^*_{ij} - \max_{j' \neq j} W^*_{ij'}.$$

Genes were ranked in descending order of s_{ij}. This ranking defines both the top-n_{top} genes used to score samples (Supplementary Data: factor-specific gene lists) and, for the top 50 genes per factor, the loadings used to quantify factor–program correspondence.

For each factor j and each reference gene program G_k, we computed the Spearman rank correlation between the continuous factor loadings $W_{\cdot j}$ (restricted to the union of the per-factor top-50 gene sets intersected with the reference programs) and a binary indicator vector $\mathbf{1}[\text{gene} \in G_k]$ (main-text Fig. 2a,b; Methods). P values were derived from the asymptotic Spearman test and adjusted for multiple comparisons using the Benjamini–Hochberg procedure within each factor column independently. Reference programs were included in the visualization only if at least one factor yielded a Spearman correlation exceeding 0.2.

Reference gene programs displayed spanned classical and basal-like tumor signatures and activated/immune stromal programs from Moffitt et al.
[14]; classical and exocrine subtypes from Collisson et al.
[19]; classical-tumor, basal-like-tumor, exocrine, and activated-stroma compartment signatures from DECODER[20]; the pure classical, immune classical, desmoplastic, and basal-like tumor microenvironment programs from Puleo et al. [16]; the ECM and immune programs from Maurer et al. [21]; the iCAF cancer-associated fibroblast signature from Elyada et al. [22]; the proCAF, restCAF, and panCAF fibroblast subtypes from SCISSORS[23]; the proCAF and restCAF programs from the DeCAF study[18]; the classical A subtype from Chan-Seng-Yue et al.
[24]; and the ADEX (aberrantly differentiated endocrine exocrine) program from Bailey et al. [15]. Bailey's ADEX program is retained because it is unique, whereas redundant Bailey programs, periductal subtypes, and MSI subtypes were dropped as exact duplicates of, or low-correlation relative to, other retained programs.

9 Reconstruction and Survival Contribution per Factor

NMF does not provide an orthogonal variance decomposition; it seeks a low-rank reconstruction of the expression matrix $X \approx WH$. We therefore measure each factor's contribution to reconstruction rather than to "variance explained.'' For a subset of factors $S \subseteq \{1, \ldots, k\}$, define the reconstruction value

$$v(S) = \|X\|_F^2 - \left\| X - \sum_{j \in S} W_{\cdot j} H_{j \cdot} \right\|_F^2,$$

the reduction in residual error relative to the null (zero-factor) model, where $W_{\cdot j} H_{j \cdot}$ is the rank-one outer-product contribution of factor j (the outer product of column j of W with row j of H). The reconstruction Shapley value[25] of factor j,

$$\phi_j = \sum_{S \subseteq \{1, \ldots, k\} \setminus \{j\}} \frac{|S|!\,(k - |S| - 1)!}{k!} \big(v(S \cup \{j\}) - v(S) \big),$$

is its average marginal reconstruction gain over all orderings of the factors. By the Shapley efficiency axiom, $\sum_j \phi_j = v(\{1, \ldots, k\})$, the total reconstruction gain of the full model, so the normalized shares $\phi_j / \sum_l \phi_l$ form an additive decomposition of reconstruction that sums to one; individual shares may be signed in principle but were all positive in the reported PDAC models. This is a genuine additive decomposition, in contrast to per-factor variance fractions, which for non-orthogonal factorizations such as NMF do not partition (cross-terms between rank-one contributions are non-zero). The Shapley shares are computed exactly by enumerating the 2^k subsets (trivial at the small ranks considered here).

The survival contribution of factor j was quantified as the Type III partial likelihood increment: the reduction in Cox partial log-likelihood when factor j is omitted from the full k-factor model,

$$\Delta\ell_j = \ell_{\text{full}}(\hat{\beta}_{\text{full}}) - \ell_{-j}(\hat{\beta}_{-j}),$$

where $\ell_{\text{full}}(\hat{\beta}_{\text{full}})$ is the maximized unpenalized Cox partial log-likelihood of the full k-factor model and $\ell_{-j}(\hat{\beta}_{-j})$ is that of the reduced model refit on the remaining $k-1$ factors after omitting factor j, each evaluated at its own maximum partial likelihood estimate. Larger $\Delta\ell_j$ indicates a greater marginal contribution of factor j to survival prediction given all other factors. Unlike the reconstruction Shapley shares, these Type III increments are conditional (each is evaluated against the full complement of the remaining $k-1$ factors) and are nonadditive: they do not partition the total model partial likelihood and are not comparable across factors as variance-style shares. We therefore report them as per-factor conditional increments rather than as an additive survival decomposition. These two quantities are plotted against each other in main text, Fig. 2c. In PDAC at $k = 3$, the reconstruction shares are 41.8%/18.6%/39.7% for NMF (N1/N2/N3) and 18.8%/25.2%/56.0% for DeSurv (D1/D2/D3), while the survival contribution $\Delta\ell$ concentrates in DeSurv D1 ($\Delta\ell = 66.4$) with all NMF factors negligible. The factors that contribute most to reconstruction are not the most prognostic.

10 Factor Correspondence Analysis

To assess which biological programs are preserved, altered, or suppressed under survival supervision, we quantified the pairwise correspondence between NMF and DeSurv factor loadings. For each pair of factors $(j_{\text{NMF}}, j_{\text{DeSurv}})$, we computed the Spearman rank correlation between the corresponding columns of the NMF gene program matrix W_{NMF} and the DeSurv gene program matrix W_{DeSurv} over all genes. We use the full loading profile rather than a top-gene subset, which would impose an arbitrary threshold and overstate apparent preservation. Spearman correlation was used to capture monotone correspondence in gene loading ranks while remaining robust to scale differences between the two methods. Results are displayed as a heatmap with NMF factors as rows and DeSurv factors as columns (main text, Fig. 2d).

10.1 Cross-method reconstructability and reconstruction fidelity

The correlation heatmap summarizes pairwise similarity; to make a direction-aware statement of how much of each factor is *linearly reconstructable* from the other method's basis, we regressed each factor's gene-loading vector onto the full set of the other method's loading vectors and report the ordinary least-squares R^2 (Table S2). A high R^2 means the factor is well reconstructed as a linear combination of the other method's factors; a low R^2 means it is not linearly reconstructable from that basis. This is a statement about linear reconstructability of gene-loading vectors, not about which method discovered a program first.

The headline is DeSurv D1: $R^2 = 0.09$ from the NMF basis, indicating that its gene-loading vector is not well reconstructed as a linear combination of the matched-rank NMF factors. The biologically dominant tumor (N1) and stromal (N3) programs, by contrast, are largely reconstructable from the DeSurv basis ($R^2 = 0.90$ and 0.88), so most of the unsupervised structure is retained; the principal reorganization is that the DeSurv basis resolves tumor signal into basal- and classical-aligned programs while reconstructing the exocrine factor (N2) less well ($R^2 = 0.38$).

At the whole-matrix level, this reorganization carries a real but bounded reconstruction cost (Table S3). On the conventional centered R^2, DeSurv reconstructs 0.75 of the analysis matrix against 0.81 for unsupervised NMF, a difference of 6.0 percentage points.

The reorganization therefore has a real price. On the uncentered scale used by the reconstruction helper the gap is only 1.5 percentage points, but that scale flatters any nonnegative low-rank fit to a rank matrix, and on the conventional centered definition DeSurv gives up 6.0 percentage points of reconstruction R^2 relative to unsupervised NMF (Table S3), a relative loss of about 7%. We report the centered figure in the main text.

10.2 The D1 axis is not specific to DeSurv

If the D1 tumor–stroma axis were an artifact of DeSurv's particular optimizer, independent survival-supervised methods should not recover it. We therefore fit three other supervised dimension-reduction methods on the same training cohort ($n = 273$, 139 events), namely supervised principal components (Cox-screened genes followed by PCA), one-component Cox partial least squares against martingale residuals, and a sparse Cox model (lasso-penalized, 32 selected genes), and compared their leading survival-aligned direction to the DeSurv D1 score. All three recovered D1 closely (per-patient $|r| = 0.81$ for supervised PCA, 0.83 for Cox-PLS, and 0.85 for sparse Cox; mean 0.83), and each aligned specifically with D1 rather than the stromal D2 or basal-like D3 programs (Fig. S2). By contrast, the leading *unsupervised* direction (the first principal component, a variance-dominant unsupervised direction) was only weakly correlated with D1 ($|r| = 0.10$) and instead tracked the high-variance D2/D3 stroma–basal axis. This D1 axis is thus a property of survival supervision shared across methods, consistent with its being unreconstructable from the unsupervised NMF basis (Table S2) and with standard NMF's failure to enrich for the proCAF stroma at matched rank, and not an idiosyncrasy of the DeSurv algorithm.

10.3 Robustness to tuned supervised comparators and compartment attribution

The recovery above establishes the D1 axis as a genuine survival direction shared across methods; here we ask what DeSurv adds beyond recovering it, using canonical, independently tuned comparators in place of the simple single-component versions above: supervised principal components (`superpc`; Bair–Tibshirani, with cross-validated screening threshold and component number) and penalized (lasso) Cox (`glmnet`, a single $\lambda_{\min}$ model). Cox partial least squares was not re-run in this pass, as a validated multi-component implementation was unavailable in our environment; the two canonical methods are sufficient. Every method began from the same 1,970-gene training-only expression/variance universe and performed its own feature selection; DeSurv was scored with its Bayesian-optimization-selected top-270-per-factor programs; all scoring used the frozen, rank-transformed transfer with no validation refitting.

Prediction parity was preserved: the pooled frozen-transfer C-index was 0.61 for the DeSurv linear predictor, 0.60 for supervised PCA, and 0.60 for penalized Cox. Consistent with the score-level recovery above, both tuned methods aligned with the D1 direction (training $|r| = 0.71$ for penalized Cox and 0.46 for supervised PCA) and with the DeSurv linear predictor ($|r| = 0.86$ for penalized Cox and 0.65 for supervised PCA), but were only weakly correlated with the activated-stroma/proCAF program D2 ($|r| = 0.06$ for penalized Cox and 0.14 for supervised PCA): a single supervised risk axis captures the dominant tumour-axis signal but not the separate stromal program.

The compartment attribution made this explicit. Because D1 runs from a classical-tumour, restraining-stroma protective pole to a basal-like high-risk pole, a sparse supervised risk score aligned with D1 is naturally composed of basal-like tumour markers. Accordingly, across 12 cross-validation seeds the genes selected by both methods were reproducibly enriched (hypergeometric $P < 0.01$) for the basal-like tumour program (in 12/12 seeds for penalized Cox and 12/12 for supervised PCA) and in no seed for the activated-stroma/proCAF, restraining-stroma, immune, or exocrine compartments. The supervised risk scores did nonetheless retain some stromal signal: each correlated modestly with an external activated-stroma/proCAF signature ($|r| = 0.20$ for penalized Cox and 0.23 for supervised PCA), but well below DeSurv's D2 program ($|r| = 0.55$), which tracks that signature directly. The supervised risk scores retained modest stromal signal but did not resolve it as a distinct program. DeSurv instead placed the dominant survival-associated direction within a three-program representation containing separately measurable classical/restCAF-like, proCAF-like and basal-like programs. Thus, the distinction is representational: DeSurv preserves stromal context as an explicit program even when that program does not contribute independently to the fitted linear predictor.

Proper-scoring sensitivity of the external validation

The main-text external validation reports discrimination with Harrell's C-index under a Cox proportional-hazards model. Because Harrell's C can be optimistic when hazards are non-proportional[26,27], we additionally evaluated the DeSurv linear predictor, its direction frozen from training, with a proper score, the integrated Brier score (IBS), in each validation cohort, recalibrating only its scalar slope and baseline hazard within each cohort and scoring over that cohort's follow-up. For a single time-invariant risk score under proportional hazards the monotone score-to-survival map preserves the concordance ordering, so Antolini's time-dependent concordance coincides with Harrell's C; the additive quantity is therefore the proper score, not a second concordance. Across the 5 cohorts the DeSurv predictor achieved an event-weighted integrated Brier score of 0.159 versus 0.168 for a Kaplan–Meier reference, an index of prediction accuracy ($\text{IPA} = 1 - \text{IBS}/\text{IBS}_{\text{null}}$) of 5.4% (per-cohort range 2.9–9.4%). Because each cohort's IBS is integrated over its own event-time horizon, this pooled value is a descriptive event-weighted summary rather than a single common-horizon proper score, and the per-cohort IPA range is the more directly interpretable quantity. Two caveats bound how this analysis should be read. First, the recalibration step estimates both the scalar slope and the baseline hazard from each cohort's own outcomes, so unlike the C-index and hazard-ratio analyses, which use a fully frozen predictor, this sensitivity evaluates the transported score direction under local recalibration rather than a frozen absolute-risk model. Second, the scores are computed on the same data used to fit that recalibration, with no cross-validation or resampling, so the reported IBS and IPA are apparent in-sample values and are optimistic. Both points cut against the reported improvement rather than for it, and the improvement is already small. This modest proper-score improvement over the null is consistent with the moderate discrimination (C = 0.61) and with DeSurv's contribution being a separately resolved tumour- and stroma-associated decomposition rather than higher predictive accuracy.

Proportional-hazards diagnostic

Because the pooled hazard ratio assumes proportional hazards, we tested the scaled Schoenfeld residuals of the stratified validation model ($\text{Surv}(t, \delta) \sim \text{risk\_z} + \text{strata(dataset)}$) with `survival::cox.zph`. The proportional-hazards assumption was not supported for the standardized DeSurv linear predictor ($\chi^2 = 14.7$, $\text{df} = 1$, $P = 1.3e-04$), with the score's effect attenuating over follow-up rather than reversing. The reported pooled hazard ratio should therefore be read as a time-averaged summary of a consistently prognostic score rather than a constant multiplicative effect; this time-varying behavior is precisely why we additionally report the horizon-integrated proper-scoring (IBS/IPA) sensitivity above, which integrates prediction accuracy over each cohort's event-time horizon (to the 80th percentile of event times) rather than through a single hazard-ratio summary.

Independence of discovery and validation

To make the separation between discovery, adjustment, and validation unambiguous: DeSurv is trained with survival supervision only; it never uses PurIST, DeCAF, or any molecular subtype label during factorization or hyperparameter selection. The PurIST and DeCAF subtype calls were fixed before the DeSurv analysis (loaded as precomputed per-cohort annotations) and enter only as covariates in the adjusted Cox models, not as inputs to DeSurv. All rank and hyperparameter selection (including the one-standard-error rule) was performed by cross-validation entirely within the TCGA and CPTAC training data; no external validation cohort was used for training or tuning. Validation-cohort expression enters only at scoring, a fixed-basis projection ($Z = \widetilde{W}^{\top} X_{\text{new}}$) that uses no validation outcome, and validation-cohort survival outcomes were used only for the final, out-of-sample performance evaluation.

Variance partition of the D1 score

The main text reports the fraction of the D1 score's variance explained by the binary PurIST and DeCAF classifications (`code/14`). D1 scores are first standardized within cohort, and an ordinary least-squares model is then fitted with cohort as a fixed effect together with PurIST, DeCAF and their interaction. The quantity reported is the ordinary R^2 of that model, not the adjusted R^2; on $n = 570$ with eight estimated parameters the adjusted value is about one percentage point lower. Because the scores are standardized within cohort before fitting, the cohort main effect absorbs essentially no variance, so the within-cohort normalization of the reported statistic is a formality rather than a correction. The interaction term contributes negligibly, so the additive and interaction models give the same figure to the reported precision. Prognostic increments over the classifiers are assessed separately, by a partial likelihood-ratio test and the change in concordance from adding D1 to a cohort-stratified Cox model containing both classifiers.

A stronger version of the same question is whether D1 is merely a smoothed coding of the four joint PurIST-by-DeCAF subtype cells, so that the classical and restCAF-like arms of D1 would simply re-express one cell of a two-by-two. We tested this directly by adding D1 to a cohort-stratified Cox model that already contains the PurIST-by-DeCAF interaction (`code/14`). The interaction term itself is not prognostic (HR 1.42, $P = 0.28$), so the joint classification adds nothing over the additive one. D1 does: adding it gives a partial likelihood-ratio statistic of 10.44 on 1 degree of freedom ($P = 0.0012$), an adjusted hazard ratio per standard deviation of 0.82 (0.73–0.92), and a concordance increase from 0.592 to 0.621. Re-coding the two classifiers as a single saturated four-level categorical variable is the same model reparameterized and returns the identical statistic ($\chi^2 = 10.44$). D1 is therefore not a relabeling of the joint subtype classification.

Adjustment for clinical covariates and tumor purity

The DeSurv linear predictor was refitted in each cohort before and after adjustment for the clinical covariates that cohort annotates and, where available, for tumor purity (Table S4). Purity annotation is present in only a subset of cohorts and is derived by different estimators across them, so it is used here solely as an adjustment covariate; wherever purity could be measured on a full cohort the adjusted hazard ratio was essentially unchanged from the unadjusted one, which argues against the association being driven by tumor purity or composition.

Part II

Supplementary Results

11 Empirical optimization stability across random initializations

The idealized alternating scheme is not accompanied by a formal convergence guarantee for the fitted models, both because the reported analyses set the loadings ridge penalty to zero ($\lambda_H = 0$) and because the implemented algorithm combines column normalization, gradient balancing between the reconstruction and Cox terms, and a Coxnet approximation of the β update. We therefore assessed optimization behavior empirically across $R = 100$ independent random initializations of the selected PDAC model, with a stopping tolerance of 10^{-7} and a maximum of 4,000 iterations. All 100 runs terminated on the stopping criterion rather than the iteration cap, none failed numerically, and in every run the recorded penalized objective decreased monotonically to a numerically stable value; the median number of iterations was 38 (range 17–233). The size of the decrease from a random start varies widely between initializations, but that quantity reflects how far a given random start sat from a local optimum rather than whether the runs agree at the end. The question that matters for this paper, whether independent runs of the reported procedure recover the same gene programs, is addressed directly in Stability of the recovered gene programs below. These analyses document empirical optimization behavior for the reported application rather than providing a formal convergence guarantee.

12 Stability of the recovered gene programs

The previous section documents that the optimizer converges. It does not establish that the programs it converges to are the same ones each time, and that is the property the manuscript relies on when it applies a frozen decomposition across cohorts. Because we raise the non-identifiability of reconstruction-based factorization as a limitation of unsupervised approaches, the same standard should be applied here. We therefore measured stability at the level of the recovered programs rather than the objective, using the same $R = 100$ cached initializations (`code/20`, `code/21`). No model was refitted for the restart analysis.

Individual restarts are highly dispersed. Comparing pairs of independent random restarts, and matching their columns one to one by absolute Spearman correlation of gene loadings, the median overlap between the top 270 genes of matched programs was 0.13 by Jaccard index, against a chance baseline of 0.074 for two independent draws of that size from 1,970 genes. Matched gene-loading correlations were moderate (0.59–0.60 across the three programs). Over the same runs, training concordance was nearly constant (median 0.704, standard deviation 0.010). Near-identical predictive performance from bases that share little gene membership is the signature of a non-identifiable objective, and it is precisely why the reported model is not a single run.

The reported procedure is reproducible. The reported basis is obtained by aggregating restarts into a consensus initialization and then fitting from it. To test whether that procedure returns the same programs, we partitioned the cached restarts into two disjoint halves, built a consensus initialization from each half independently, fitted from each at the production settings, and compared the results (8 random splits). Two independently seeded consensus fits agreed closely with each other (gene-loading correlation 0.93–0.95, patient-score correlation 0.97–0.99), and each agreed still more closely with the reported basis (gene-loading correlation 0.97–0.99, top-270 Jaccard 0.73–0.88). Splitting the restarts halves the number aggregated relative to the reported procedure, so these figures are conservative.

The two results should be read together. The optimizer alone is not identifiable, and the consensus step is what makes the reported decomposition a stable object rather than one draw from a wide distribution. Reproducibility claims in this manuscript refer to that end-to-end procedure.

Stability under data perturbation is a separate and weaker property. The analyses above vary the initialization while holding the data fixed. Refitting on repeated 80% subsamples of the training cohort and comparing top-100 gene sets instead gives a mean pairwise Jaccard of 0.34 for DeSurv, against 0.49 for Cox partial least squares, 0.38 for supervised principal components and 0.22 for sparse Cox. DeSurv is mid-range. This comparison favours dense methods by construction, because a score that spreads weight across many genes has a more stable top-100 list than one built from a selective basis, as the sparse Cox value illustrates. We report it because gene-level resampling stability is a reasonable thing for a reader to ask about, not because it is the property the frozen-program argument rests on.

13 Simulation results under null and mixed scenarios

To characterize the operating characteristics of DeSurv across different signal regimes, we evaluated performance under three simulation scenarios. In the prognostic scenario (main text Fig. 5), prognostic programs explained low variance relative to outcome-neutral background signals, and survival depended on 150 factor-specific marker genes ($\beta_1 = 2$, $\beta_2 = \beta_3 = 0$). In the null scenario, survival times were generated independently of gene expression ($\beta = 0$), so no prognostic structure existed. In the mixed scenario, survival depended on 300 genes per lethal factor, of which 50% were factor-specific markers and 50% were background genes with loadings shared across all factors. All scenarios used $G = 3{,}000$ genes, $N = 200$ samples, $K = 3$ true factors, and identical gamma distribution parameters for the gene loading matrix W (see Methods). Performance was evaluated using cross-validated concordance index (C-index), precision of recovered prognostic gene sets, and the distribution of selected ranks across simulation replicates.

Under the null scenario (Fig. S3a–b), DeSurv showed no advantage over unsupervised NMF ($\alpha = 0$): both methods yielded C-index values near 0.5, and selected ranks concentrated at low values (NMF favoring $k = 2$), consistent with the absence of survival signal. DeSurv therefore did not recover spurious prognostic structure when none existed. Because the survival objective is uninformative under the null, the cross-validated concordance is essentially flat in α and the Bayesian-optimized supervision strength is unidentified, dispersing across its admissible range rather than concentrating near zero (median $\alpha = 0.59$, IQR 0.19–0.82; range 0–0.94 across 100 replicates). The null-scenario specificity thus reflects the flat concordance surface itself, not the optimizer driving α to zero.

Under the mixed scenario (Fig. S3c–f), DeSurv's advantage was concentrated in marker-gene recovery. Precision against the factor-specific marker genes alone (150 genes; Fig. S3d left) substantially favored DeSurv, whereas precision against the full 300-gene survival set including background genes (Fig. S3d right) was more modestly improved, indicating similar background-gene recovery between methods. The learned factor aligned with the true marker program also received a larger Cox coefficient ($|\beta|$) under DeSurv (Fig. S3e), and C-index was modestly higher (Fig. S3c). Across the three scenarios the pattern was graded: DeSurv's advantage was greatest when variance and prognosis diverged (prognostic), attenuated when they partially overlapped (mixed, as background-gene dilution reduced the discrimination gain), and absent when no survival signal existed (null). DeSurv is therefore most useful when the dominant transcriptional axes are expected to be prognostically neutral, as in cancers with high tumor-purity variation or dominant tissue-composition signals.

14 Cross-validated C-index across factorization rank

To evaluate how the number of latent factors k affects prognostic performance, as a sensitivity analysis we performed an exhaustive grid search over $k \in \{2, 3, \ldots, 12\}$ and supervision strength $\alpha \in \{0, 0.05, \ldots, 0.95\}$ using 5-fold cross-validation on the TCGA+CPTAC training cohort. For each k, the best-performing α was selected by maximizing the mean CV C-index. The unsupervised NMF baseline ($\alpha = 0$) was evaluated at the same k values for comparison. Results are shown for two gene-selection settings: $n_{\text{top}} = 270$ top genes per factor and $n_{\text{top}} = \text{ALL}$ (all genes retained), both at $\lambda = 0.349$ and $\xi = 0.056$. External validation C-index is shown descriptively; validation outcomes were not used to select k, α, λ, ξ, or n_{top}.

Fig. S4 shows the resulting C-index curves for both training (CV) and external validation (pooled across held-out cohorts), with shaded ribbons indicating ± 1 standard error. Rank was selected on the training cross-validated concordance: for DeSurv the curve is close to flat across the whole range examined, spanning about two standard errors from $k = 2$ to $k = 12$ with its nominal maximum at $k = 9$ rather than at the selected rank, and the one-standard-error rule accordingly selected the parsimonious $k = 3$. The selection depends on the gene-selection setting: under panel (b), with all genes retained, the same rule selects $k = 2$. DeSurv exceeded unsupervised NMF at matched rank at every k examined. The external-validation curve is shown only descriptively (validation outcomes were used for no part of rank or hyperparameter selection) and is consistent with this training-based choice, with DeSurv at $k = 3$ attaining concordance comparable to standard NMF at higher ranks (e.g., $k = 7$; Table S5). We therefore read this as a training/cross-validation sensitivity analysis of rank, not as an external model-selection argument.

15 Standard NMF rank selection heuristics

Standard unsupervised rank selection criteria (reconstruction residuals, cophenetic correlation coefficient, and mean silhouette width) yielded inconsistent guidance for selecting the factorization rank k in the PDAC training cohort (TCGA + CPTAC). Reconstruction residuals decreased smoothly without a clear elbow (Fig. S5a). Cophenetic correlation fluctuated without a distinct transition point (Fig. S5b). Mean silhouette width favored very low ranks that conflicted with other criteria (Fig. S5c). These contradictions motivated the survival-supervised rank selection approach described in the main text.

16 Standard NMF factor structure at independently selected rank

Throughout the main text, both DeSurv and standard NMF were fit at the same rank ($k = 3$), enabling a controlled comparison of the effect of survival supervision on factor structure. When standard NMF selects its own rank independently, Bayesian optimization of the unsupervised ($\alpha = 0$) baseline selects $k = 7$, whereas supervised DeSurv BO selects $k = 3$. We examine here the NMF factor structure at this independently selected $k = 7$ to illustrate the additional fragmentation that arises when supervision is absent (Fig. S6).

At BO-selected $k = 7$ under $\alpha = 0$ (Fig. S6), standard NMF shows increased fragmentation relative to the matched-rank $k = 3$ solution: multiple factors partially overlap the same PDAC gene signatures, and no additional factor maps cleanly onto a distinct established PDAC program; the added factors fragment existing signatures rather than resolving new biology.

Stromal resolution: which reference programs are recovered from bulk

Bulk deconvolution has conventionally separated activated from normal stroma, whereas the finer cancer-associated fibroblast states, and the proCAF/restCAF classification derived from them, were established in single-cell data. We therefore asked which stromal reference programs each $k = 3$ factorization actually resolves (`code/23`). Two views are reported because they do not agree, and the defensible claim depends on which question is asked.

In the top-gene view shown in main-text Fig. 2a,b, which asks whether signature genes are enriched among the 50 genes that define each factor, the activated-stroma program is recovered comparably by all methods (7, 6 and 10 of 206 genes for DeSurv, standard NMF and the matched-budget control). The restraining-stroma programs behave differently: 2 of 25 restCAF genes and 2 of 35 iCAF genes appear among DeSurv's defining genes, against 0 and 0 for standard NMF. The proCAF programs, by contrast, are recovered comparably by all methods (4 versus 3 of 25 genes).

In the full-gene view, which correlates loadings across all 1,970 analysed genes with signature membership, **no method separates restCAF from proCAF**: every correspondence lies between 0.04 and 0.10, whereas the activated-stroma axis is recovered by all methods at 0.19. This view is far more dilute for a 15–25 gene signature among 1,970 genes and correspondingly low powered, but it bounds what can be claimed. We therefore report the restraining-stroma difference as marker-gene presence among the defining genes of each factorization, and do not claim to have resolved the single-cell CAF states from bulk expression.

Note also that the SCISSORS iCAF and myCAF reference signatures are displayed as restCAF and proCAF in main-text Fig. 2, following the naming used by the DeCAF classification; the underlying gene sets are the single-cell-derived ones.

Matched-budget unsupervised control at the same rank

The matched-rank comparison in the main text uses Lee multiplicative NMF, which receives neither DeSurv's consensus initialization nor its hyperparameter tuning. That asymmetry leaves open the reading that the difference in factor structure reflects tuning effort rather than supervision. To close it, we fit a second unsupervised model in which every element of the DeSurv procedure is held fixed except supervision. Bayesian optimization was run with α pinned to 0 and the rank pinned to DeSurv's selected $k = 3$, using the same optimizer (`desurv_cv_bayesopt_refine`), the same search budget, the same gene filter and the same cross-validation folds; the resulting model was then fit by the same 100-restart consensus procedure used for DeSurv (`code/03` section 5, `code/04` sections 1d–1e). The control selected $\lambda = 0.0025$, $\nu = 0.0566$ and $n_{\text{top}} = 299$; note that its gene support was selected at the upper bound of the search range (300), whereas the supervised model settled at $n_{\text{top}} = 270$ well inside it. Because $\alpha = 0$ removes the survival term, the selected λ is not comparable like-for-like with the supervised model's.

The control does not recover D1. Its best-matching factor reaches Spearman $\rho = 0.24$ against D1, slightly lower than the $\rho = 0.36$ attained by `NMF::nmf` at the same rank, while still recovering D2 ($\rho = 0.86$) and D3 ($\rho = 0.75$). Its external discrimination is likewise no better than `NMF::nmf` at $k = 3$ (Table S5). A sensitivity analysis holding the supervised model's own λ, ν and n_{top} fixed and varying only α gave the same conclusion. Matching the tuning budget therefore does not close the representational gap, and the reorganization reported in the main text is attributable to supervision rather than to computational effort.

Standard NMF at elbow-selected $k = 5$ and BO-selected $k = 7$ both substantially improve over NMF at $k = 3$ across all cohorts (Table S5); additional unsupervised factors capture prognostically relevant variation missed at the matched rank. At $k = 7$, NMF matches or exceeds DeSurv's per-cohort C-index in most cohorts, indicating that unsupervised factorization can approach supervised concordance given enough factors. This comes at the cost of a less parsimonious and less interpretable factorization: the $k = 7$ NMF solution shows increased fragmentation (Fig. S6), distributing the same prognostic signal across a rank-sensitive set of overlapping factors rather than concentrating it. The DeSurv linear predictor, by contrast, retains significant prognostic value after adjustment for established classifiers with only three factors (main-text Table 1).

Supplementary Data: factor-specific gene lists

As Supplementary Data, Tables S6–S8 list the top 270 genes for each of the three DeSurv factors, ranked by the factor specificity score s_{ij} defined in Supplementary Information, Section 8. The 270-gene cutoff corresponds to the Bayesian optimization–selected projection dimension ($n_{\text{top}} = 270$); these are the genes used to compute sample-level factor scores for survival prediction. Factor D1 is enriched for classical tumor and restCAF-like stromal genes, Factor D2 for proCAF-associated stromal genes, and Factor D3 for basal-like tumor genes.

17 Translational transportability into independently treated PDAC cohorts

The programs were learned in treatment-naive, non-metastatic PDAC, the setting in which the gene-program matrix W and Cox coefficients β were trained. As a translational test we projected the fixed programs, without retraining, into independently treated PDAC and asked whether they retained clinically relevant associations. We used two resources. First, the O'Kane/COMPASS cohort of treated metastatic PDAC (laser-capture microdissected and therefore epithelial-enriched), which supports a transportability test of the tumor-associated D_1 direction under a model stratified by treatment arm (Fig. 4a). Second, the borderline-resectable / locally advanced Linehan cohort, in which paired pre- and post-treatment biopsies allow a within-patient test of whether the stromal D_2 program changes with therapy (Fig. 4b). Both are analysed within a single cohort and, for O'Kane/COMPASS, within treatment arm; no estimate is pooled across cohorts or across treatment arms. The elastic-net penalty shrank the trained survival coefficient for D_2 to exactly zero in the treatment-naive model (an estimated selection outcome, not an imposed constraint), so the paired analysis is reported as a change in program level and not as evidence that D_2 predicts survival.

17.1 Prognostic transportability of D1 in treated metastatic disease

To evaluate transportability in treated metastatic disease without combining treatment arms, we projected the frozen DeSurv programs into the O'Kane/COMPASS cohort and estimated D_1 associations separately within each arm. D_1 was associated with lower mortality in the same direction in FOLFIRINOX (HR per SD 0.80 (0.61–1.03)), gemcitabine/nab-paclitaxel (HR 0.57 (0.37–0.86)) and the GA/experimental arm (HR 0.83 (0.58–1.20); Fig. 4a). An arm-stratified Cox model allowing treatment-specific baseline hazards yielded a common association of HR 0.75 (0.62–0.90) ($P = 0.003$; 173 patients, 123 deaths), with no evidence of heterogeneity across treatment arms (interaction $P = 0.40$). Because these specimens were epithelial-enriched by laser-capture microdissection, this analysis specifically tests transportability of the tumor-associated D_1 direction rather than the stromal programs.

17.2 Treatment-associated change in D2

In the 29 Linehan patients with paired pre- and post-treatment biopsies, D_2 increased significantly from pre- to post-treatment (paired Wilcoxon 0.013; mean $\Delta = +0.34$ SD; Fig. 4b), consistent with treatment-associated remodeling of the proCAF program.

17.3 Caveats

These analyses are exploratory and prognostic, not predictive: treatment, stage and cohort are confounded, the treated cohorts are modestly sized, and no analysis plan was preregistered. Each analysis is confined to a single cohort, and the O'Kane/COMPASS estimate is stratified by treatment arm; nothing is pooled across cohorts or across arms. We report no program-by-survival association in bulk treated tissue. Robust treated-cohort validation, and any assessment of whether the programs predict differential treatment benefit, will require larger harmonized series with adequately powered treatment-by-program interaction analyses.

SI References

1. Lee, D. D. & Seung, H. S. Algorithms for non-negative matrix factorization. in *Advances in neural information processing systems 13 (NIPS 2000)* (eds. Leen, T. K., Dietterich, T. G. & Tresp, V.) 556–562 (MIT Press, 2001).
2. Pascual-Montano, A., Carazo, J. M., Kochi, K., Lehmann, D. & Pascual-Marqui, R. D. Nonsmooth nonnegative matrix factorization (nsNMF). *IEEE Transactions on Pattern Analysis and Machine Intelligence* **28**, 403–415 (2006).
3. Simon, N., Friedman, J. H., Hastie, T. & Tibshirani, R. Regularization paths for cox's proportional hazards model via coordinate descent. *Journal of Statistical Software* **39**, 1–13 (2011).

4. Donoho, D. & Stodden, V. When does non-negative matrix factorization give a correct decomposition into parts? in *Advances in neural information processing systems 16 (NIPS 2003)* (eds. Thrun, S., Saul, L. K. & Schölkopf, B.) 1141–1148 (MIT Press, 2004).

5. Laurberg, H., Christensen, M. G., Plumbley, M. D., Hansen, L. K. & Jensen, S. H. Theorems on positive data: On the uniqueness of NMF. *Computational Intelligence and Neuroscience* **2008**, 764206 (2008).

6. Gillis, N. The why and how of nonnegative matrix factorization. in *Regularization, optimization, kernels, and support vector machines* (eds. Suykens, J. A. K., Signoretto, M. & Argyriou, A.) 257–291 (Chapman; Hall/CRC, 2014).

7. Brunet, J.-P., Tamayo, P., Golub, T. R. & Mesirov, J. P. Metagenes and molecular pattern discovery using matrix factorization. *Proceedings of the National Academy of Sciences USA* **101**, 4164–4169 (2004).

8. Kim, J. & Park, H. *Sparse Nonnegative Matrix Factorization for Clustering.* https://faculty.cc.gatech.edu/~hpark/papers/GT-CSE-08-01.pdf (2008).

9. Gillis, N. & Glineur, F. Accelerated multiplicative updates and hierarchical ALS algorithms for nonnegative matrix factorization. *Neural Computation* **24**, 1085–1105 (2012).

10. Blei, D. M. & McAuliffe, J. D. Supervised topic models. in *Advances in neural information processing systems 20 (NIPS 2007)* (eds. Platt, J. C., Koller, D., Singer, Y. & Roweis, S. T.) 121–128 (Curran Associates, Inc., 2007).

11. Raphael, B. J. *et al.* Integrated genomic characterization of pancreatic ductal adenocarcinoma. *Cancer Cell* **32**, 185–203 (2017).

12. Cao, L. *et al.* Proteogenomic characterization of pancreatic ductal adenocarcinoma. *Cell* **184**, 5031–5052 (2021).

13. Dijk, F. *et al.* Unsupervised class discovery in pancreatic ductal adenocarcinoma reveals cell-intrinsic mesenchymal features and high concordance between existing classification systems. *Scientific Reports* **10**, 337 (2020).

14. Moffitt, R. A. *et al.* Virtual microdissection identifies distinct tumor- and stroma-specific subtypes of pancreatic ductal adenocarcinoma. *Nature Genetics* **47**, 1168–1178 (2015).

15. Bailey, P. *et al.* Genomic analyses identify molecular subtypes of pancreatic cancer. *Nature* **531**, 47–52 (2016).

16. Puleo, F. *et al.* Stratification of pancreatic ductal adenocarcinomas based on tumor and microenvironment features. *Gastroenterology* **155**, 1999–2013 (2018).

17. Rashid, N. U. *et al.* Purity independent subtyping of tumors (PurIST), a clinically robust, single-sample classifier for tumor subtyping in pancreatic cancer. *Clinical Cancer Research* **26**, 82–92 (2020).

18. Peng, X. L. *et al.* DeCAF defines clinical fibroblast subtypes and multidimensional tumor-stroma crosstalk shaping prognosis and immunotherapy response. *Cell Reports Medicine* **7**, 102611 (2026).

19. Collisson, E. A. *et al.* Subtypes of pancreatic ductal adenocarcinoma and their differing responses to therapy. *Nature Medicine* **17**, 500–503 (2011).

20. Peng, X. L., Moffitt, R. A., Torphy, R. J., Volmar, K. E. & Yeh, J. J. De novo compartment deconvolution and weight estimation of tumor samples using DECODER. *Nature Communications* **10**, 4729 (2019).

21. Maurer, C. *et al.* Experimental microdissection enables functional harmonisation of pancreatic cancer subtypes. *Gut* **68**, 1034–1043 (2019).

22. Elyada, E. *et al.* Cross-species single-cell analysis of pancreatic ductal adenocarcinoma reveals antigen-presenting cancer-associated fibroblasts. *Cancer Discovery* **9**, 1102–1123 (2019).

23. Leary, J. R. *et al.* Sub-cluster identification through semi-supervised optimization of rare-cell silhouettes (SCISSORS) in single-cell RNA-sequencing. *Bioinformatics* **39**, btad449 (2023).

24. Chan-Seng-Yue, M. *et al.* Transcription phenotypes of pancreatic cancer are driven by genomic events during tumor evolution. *Nature Genetics* **52**, 231–240 (2020).
25. Shapley, L. S. A value for n-person games. in *Contributions to the theory of games, volume II* (eds. Kuhn, H. W. & Tucker, A. W.) 307–317 (Princeton University Press, 1953).
26. Birolo, G. *et al.* Beyond Cox models: Assessing the performance of machine-learning methods in non-proportional hazards and non-linear survival analysis. *Computers in Biology and Medicine* **198**, 111176 (2025).
27. Hao, L., Kim, J., Kwon, S. & Ha, I. D. Deep learning-based survival analysis for high-dimensional survival data. *Mathematics* **9**, 1244 (2021).

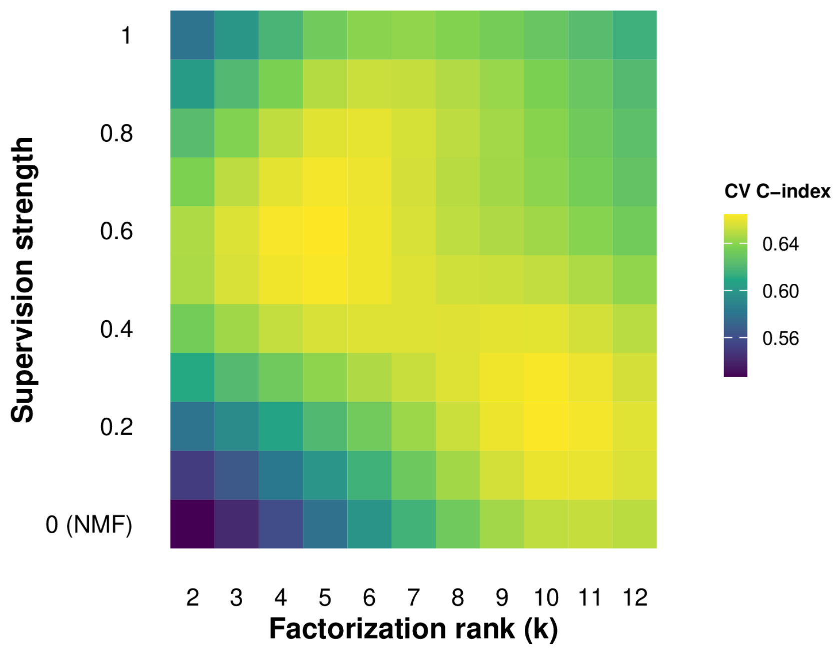

Figure S1: Bayesian-optimization tuning surface for DeSurv in the PDAC training cohort (TCGA + CPTAC). Gaussian-process–predicted mean cross-validated C-index over factorization rank (k) and supervision strength (α; $\alpha = 0$ corresponds to standard NMF). The response surface shows a broad region of high cross-validated performance at intermediate supervision and moderate rank, with discrimination degrading toward the unsupervised ($\alpha = 0$) boundary. This is empirical model-selection tuning on the training data, not a simulation result.

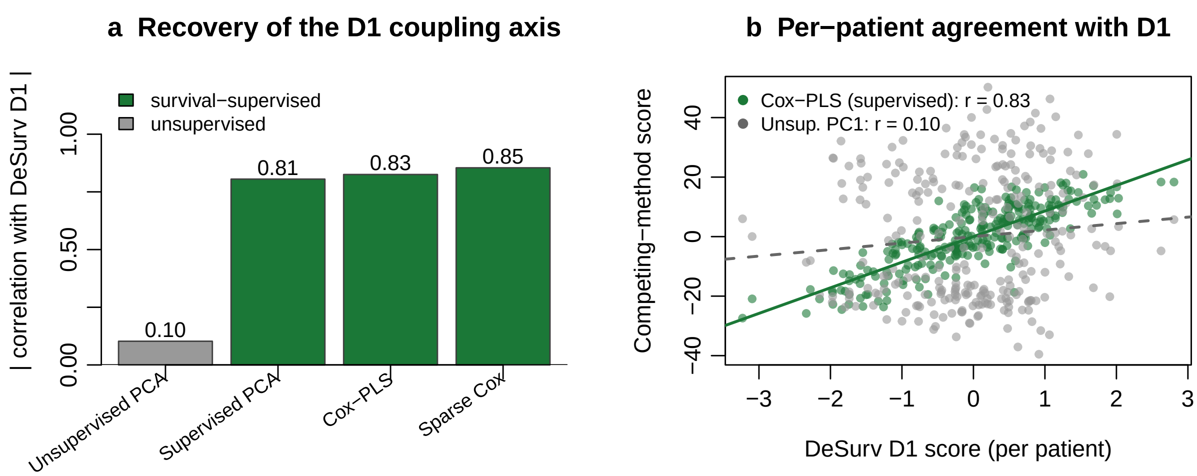

Figure S2: Independent survival-supervised methods recover the D1 axis; an unsupervised method does not. (a) Absolute correlation (across training-cohort patients) of each method's leading direction with the DeSurv D1 score, in the training cohort: supervised principal components, Cox partial least squares, and sparse Cox all recover D1 (green), whereas the leading unsupervised principal component does not (grey). (b) Sample-level agreement with D1 for Cox-PLS (supervised, green) versus the first unsupervised principal component (grey).

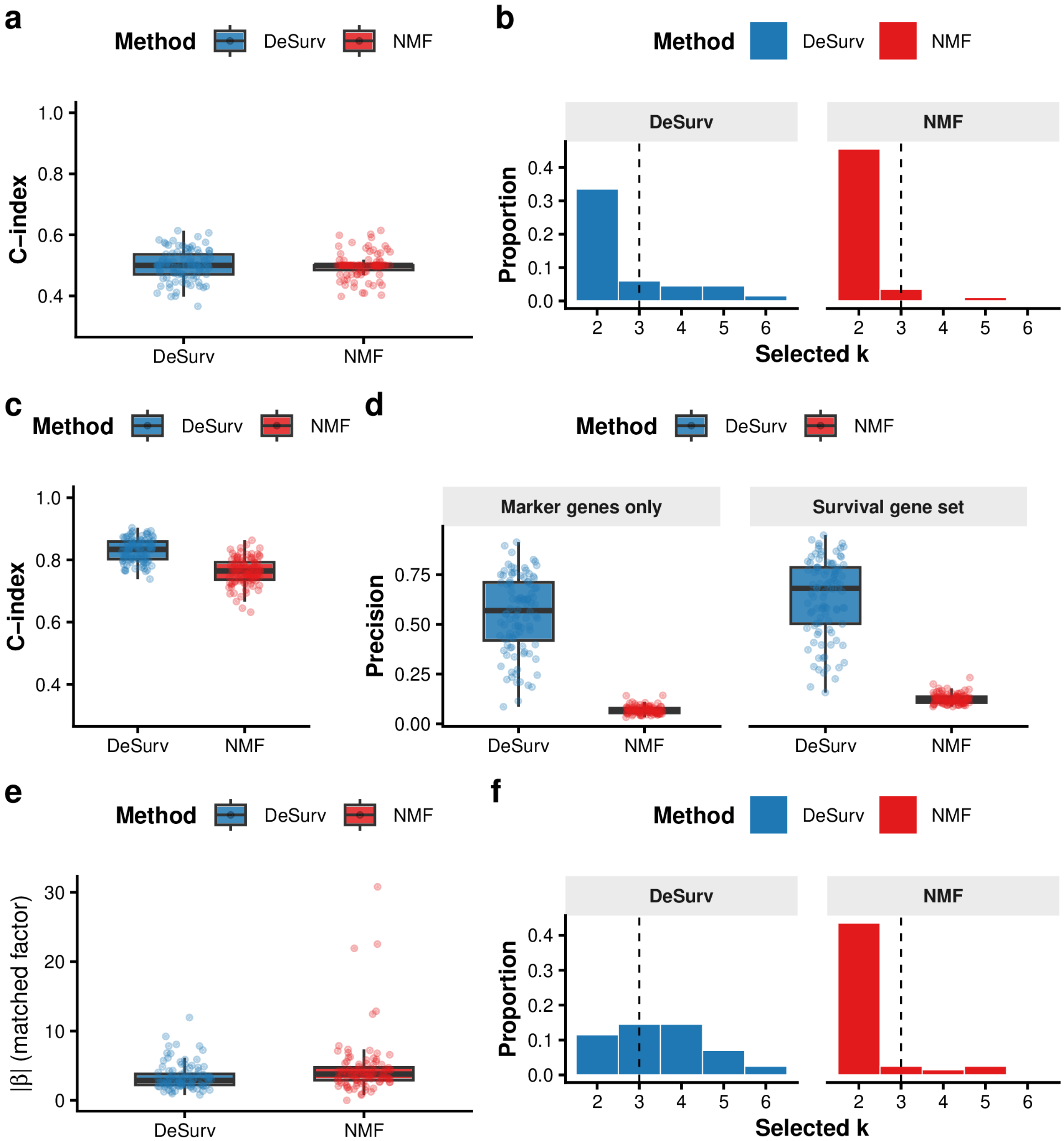

Figure S3: DeSurv performance under null and mixed simulation scenarios. (a–b) Null scenario ($\beta = 0$): both methods yield C-index ≈ 0.5 (a) and comparable rank distributions (b), confirming DeSurv does not recover spurious prognostic structure; precision is omitted as no true survival genes exist. (c–f) Mixed scenario: survival depends on factor-specific marker genes (50%) and shared background genes (50%). (c) C-index is modestly higher for DeSurv. (d) Precision by gene-set definition: marker-only precision (150 genes; left facet) substantially favors DeSurv; full 300-gene survival-set precision (right facet) shows a more modest improvement. (e) The matched factor receives a substantially larger Cox coefficient $|\beta|$ under DeSurv, confirming preferential amplification of marker-specific signal. (f) Selected-rank distributions. DeSurv (blue); NMF (red, $\alpha = 0$); both Bayesian-optimized.

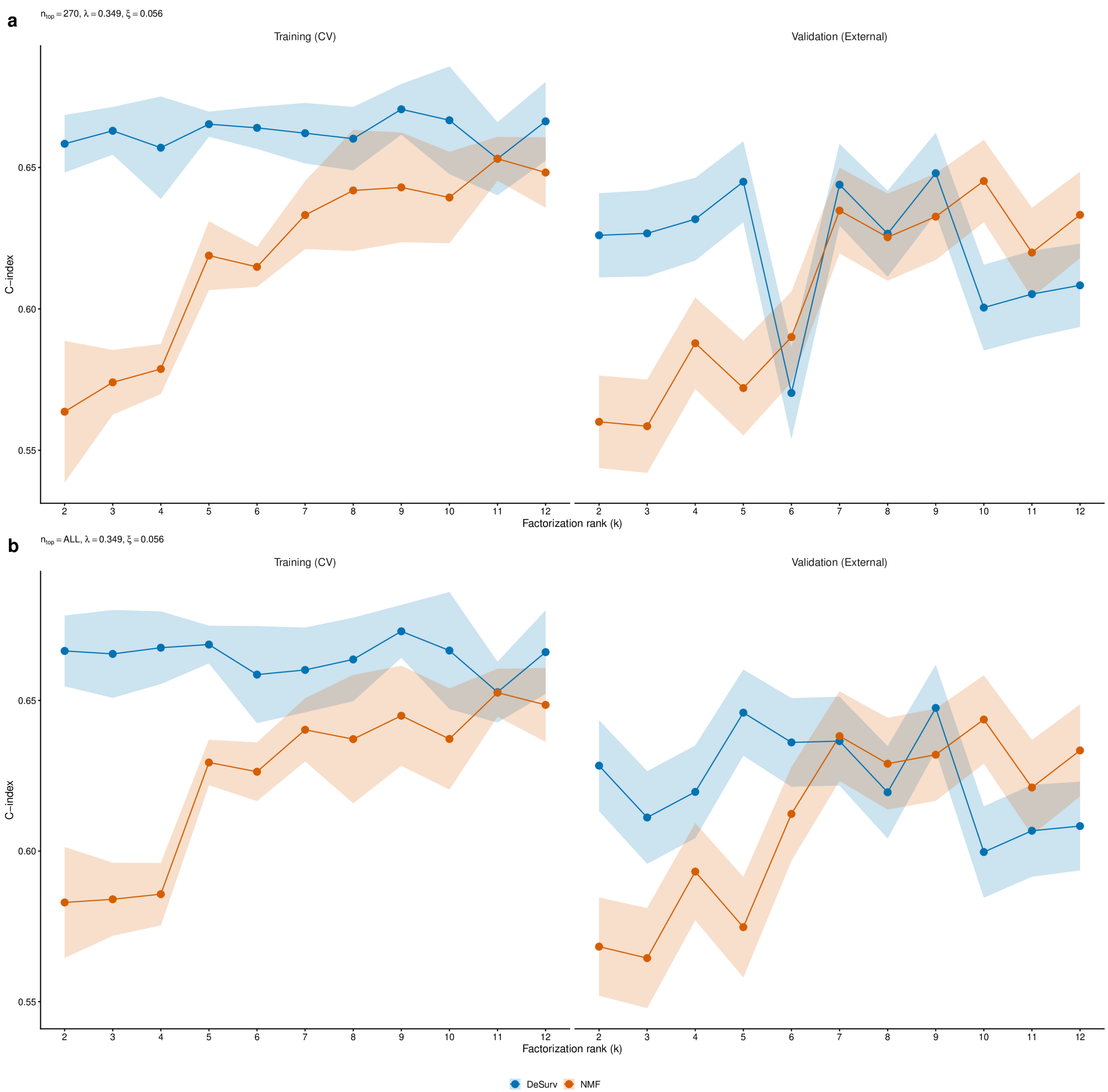

Figure S4: Training (CV) and external validation C-index as a function of factorization rank k for DeSurv (blue) and standard NMF (red, $\alpha = 0$). Shaded ribbons show ± 1 SE. For each k, the best α was selected by maximizing the mean CV C-index. Validation C-index is pooled across all external cohorts. (a) Results using $n_{\text{top}} = 270$ top genes per factor ($\lambda = 0.349$, $\xi = 0.056$). (b) Results retaining all genes ($n_{\text{top}} = \text{ALL}$; $\lambda = 0.349$, $\xi = 0.056$).

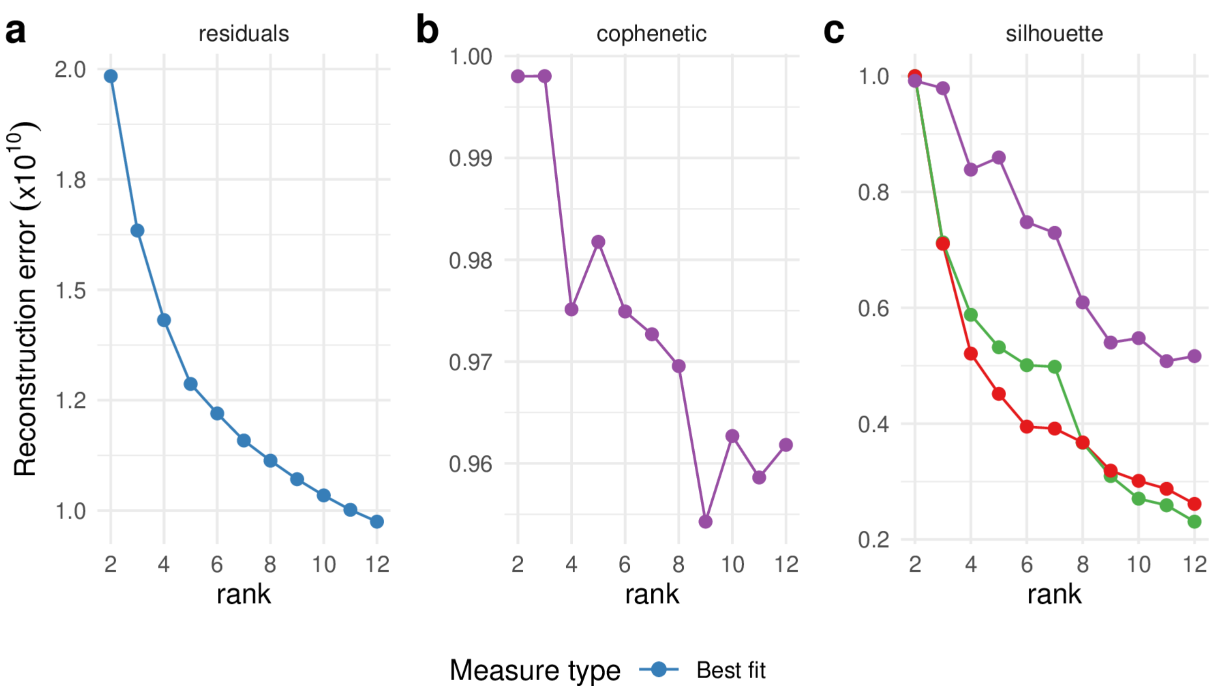

Figure S5: Standard unsupervised NMF rank selection heuristics for the PDAC training cohort (TCGA + CPTAC). (a) Reconstruction residuals as a function of factorization rank k. (b) Cophenetic correlation coefficient as a function of k. (c) Mean silhouette width across multiple distance metrics as a function of k. These criteria yield inconsistent guidance, motivating the survival-supervised model-selection approach described in the main text.

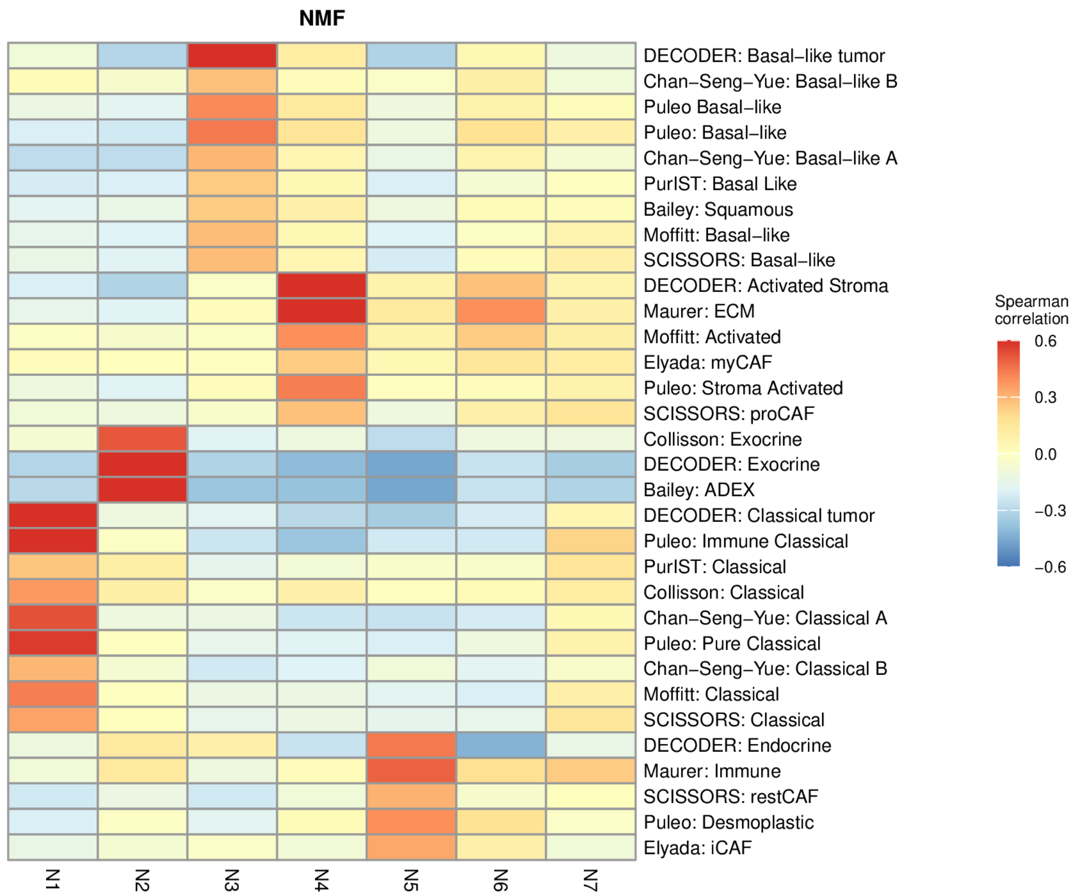

Figure S6: Gene-program correspondence for standard NMF at $k = 7$ (BO-selected rank with $\alpha = 0$), computed with the same Spearman-correlation analysis and reference programs as main-text Fig. 2a,b (top 50 genes per factor; reference programs with $r > 0.2$ for at least one factor are shown). Compared with the matched-rank $k = 3$ solution, the 7-factor solution shows increased fragmentation: multiple factors correlate with the same established PDAC signatures and no additional factor maps cleanly onto a distinct program.

Table S1: Summary of PDAC cohorts used in this study. N denotes the number of samples after all quality-control filters (histology, survival validity, dataset-specific exclusions). Patient-level metadata: OS = overall survival (time and event indicator); PurIST = tumor subtype classification (Rashid et al. 2020); DeCAF = CAF subtype classification (Peng et al. 2026). Expression data and molecular classifications for the validation cohorts were originally compiled for the DeCAF study (Peng et al. 2026). Events denotes the number of overall-survival events (deaths) with the corresponding event rate; outcome frequency was lower and more homogeneous in the training cohorts (50–52%) than across the validation cohorts (60–90%, pooled 67%), a case-mix difference relevant to the interpretation of external-validation performance.

Cohort	Platform	N	Events	Role	Accession	Metadata	Reference
TCGA-PAAD	RNA-seq	144	75 (52%)	Training	GDC (TCGA-PAAD)	OS, PurIST, DeCAF	Raphael et al. (2017)
CPTAC	RNA-seq	129	64 (50%)	Training	GDC (CPTAC-3)	OS, PurIST, DeCAF	Cao et al. (2021)
Dijk	RNA-seq	90	81 (90%)	Validation	E-MTAB-6830	OS, PurIST, DeCAF	Dijk et al. (2020)
Moffitt	Microarray	123	83 (67%)	Validation	GSE71729	OS, PurIST, DeCAF	Moffitt et al. (2015)
PACA-AU (array)	Microarray	63	38 (60%)	Validation	GSE36924	OS, PurIST, DeCAF	Bailey et al. (2016)
PACA-AU (RNA-seq)	RNA-seq	52	31 (60%)	Validation	EGAS00001000154	OS, PurIST, DeCAF	Bailey et al. (2016)
Puleo	Microarray	288	181 (63%)	Validation	E-MTAB-6134	OS, PurIST, DeCAF	Puleo et al. (2018)
Training total		273	139 (51%)				
Validation platform profiles total		616	414 (67%)				
Combined unique-patient validation set		570	388 (68%)				

PACA-AU contains 46 patients profiled on both the array and RNA-seq platforms; combined (pooled) analyses represent each patient once, retaining the RNA-seq profile for duplicated patients, consistent with the DeCAF preprocessing rule. Platform-specific per-dataset analyses retain both records.

Table S2: Cross-method projection R^2: how reconstructable each factor's gene-loading vector is from the other method's full basis (OLS R^2). DeSurv retains the bulk of the unsupervised structure (NMF N1 and N3 carry over at $R^2 \geq 0.88$) and reorganizes it: the survival-aligned D1 program is essentially unreconstructable from the NMF basis ($R^2 = 0.09$), i.e. not well reconstructed as a linear combination of the NMF factors, while the exocrine NMF factor N2 is poorly reconstructed from the DeSurv basis ($R^2 = 0.38$).

Factor	R^2 from other basis	Linear reconstructability
NMF N1 (tumor)	0.90	well reconstructed
NMF N2 (exocrine)	0.38	poorly reconstructed
NMF N3 (microenviron.)	0.88	well reconstructed
DeSurv D1 (classical/restCAF)	0.09	not linearly reconstructable
DeSurv D2 (proCAF)	0.81	reconstructable (≈N3)
DeSurv D3 (basal-like)	0.75	reconstructable (≈N1)

Table S3: Whole-matrix reconstruction R^2 across the 1,970-gene analysis matrix for unsupervised NMF versus DeSurv at $k = 3$. Two denominators are shown. The centered R^2 is the conventional definition, taken about the grand mean, and is the one to read as fraction of variation reconstructed. The uncentered R^2 divides by $\sum X^2$; because the analysis matrix holds within-sample ranks with grand mean $p/2$, the mean alone accounts for about 75% of that denominator, so uncentered values are high for any nonnegative low-rank fit and are reported only for comparability with the code. On the centered definition DeSurv gives up 6.0 percentage points of reconstruction relative to NMF (1.5 on the uncentered scale).

Model	Centered R^2	Uncentered R^2	Residual fraction (centered)
NMF ($k = 3$)	0.812	0.953	0.188
DeSurv ($k = 3$)	0.752	0.938	0.248

Table S4: Adjusted hazard ratios (per SD of the DeSurv linear predictor) for overall survival, before and after adjustment for standard clinical covariates and, where annotated, tumor purity, fitted separately in each cohort by complete-case analysis (per-cell n and events shown). The Covariates column lists the clinical variables entered (stage, grade, age, sex, nodal status, or resection margin, as available). After clinical adjustment the DeSurv score remains independently prognostic in Puleo, PACA-AU array and both training cohorts, and is attenuated to non-significance in PACA-AU RNA-seq, Dijk and Moffitt, the three smallest complete-case sets. Where tumor-purity annotation was available, purity adjustment left the estimate essentially unchanged wherever it could be measured on the full cohort (PACA-AU array 1.44 unadjusted versus 1.45 adjusted, PACA-AU RNA-seq 1.66 versus 1.65, TCGA 3.50 versus 3.60), which argues against the association being driven by tumor purity or composition. The exception, Moffitt, has purity for only 29 of its 123 patients. The smaller Dijk and Moffitt cohorts are individually underpowered, and their estimates are correspondingly unstable: Moffitt is non-significant both unadjusted and after adjustment, and its purity-adjusted model rests on only $n = 29$ of its 123 patients. Training-cohort (TCGA, CPTAC) hazard ratios are in-sample.

Cohort	Covariates	Unadjusted	Clinical-adjusted	Purity-adjusted
PACA-AU array	stage, grade, age, sex	1.44 (1.07–1.93), 0.016 (n=63, 38 ev)	1.58 (1.06–2.35), 0.025 (n=62, 37 ev)	1.45 (1.08–1.96), 0.014 (n=63, 38 ev)
PACA-AU RNA-seq	stage, grade, age, sex	1.66 (1.15–2.38), 0.006 (n=52, 31 ev)	1.56 (0.95–2.55), 0.077 (n=51, 30 ev)	1.65 (1.15–2.36), 0.006 (n=52, 31 ev)
Puleo	sex, margin	1.37 (1.18–1.59), < 0.001 (n=288, 181 ev)	1.33 (1.14–1.55), < 0.001 (n=284, 177 ev)	—
Dijk	grade, age, sex, nodal	1.32 (1.06–1.66), 0.015 (n=90, 81 ev)	1.18 (0.92–1.50), 0.189 (n=85, 77 ev)	—
Moffitt	sex, margin	1.20 (0.98–1.48), 0.076 (n=123, 83 ev)	1.20 (0.94–1.53), 0.154 (n=98, 67 ev)	0.87 (0.50–1.54), 0.639 (n=29, 18 ev)
TCGA (train)	stage, grade, age, sex, nodal	3.50 (2.63–4.66), < 0.001 (n=144, 75 ev)	3.32 (2.40–4.59), < 0.001 (n=142, 74 ev)	3.60 (2.68–4.84), < 0.001 (n=143, 74 ev)
CPTAC (train)	stage, age, sex, nodal	3.72 (2.73–5.08), < 0.001 (n=129, 64 ev)	3.59 (2.59–4.97), < 0.001 (n=123, 61 ev)	—

Table S5: External validation C-index by method and cohort. DeSurv at BO-selected rank ($k = 3$) is compared with standard NMF at matched rank ($k = 3$), elbow-selected rank ($k = 5$) and BO-selected rank ($k = 7$, $\alpha = 0$), and with a matched-budget unsupervised control at $k = 3$. The unsupervised columns are not all produced by one implementation: $k = 3$ (Std NMF) and $k = 5$ are Lee multiplicative NMF (`NMF::nmf`), whereas the matched-budget $k = 3$ column and the $k = 7$ column are the DeSurv optimizer run at $\alpha = 0$ with the same consensus initialization as the supervised model. The matched-budget column is the tuning-budget control: it holds rank, optimizer, consensus initialization and Bayesian-optimization budget fixed and varies only supervision, and it does not improve on `NMF::nmf` at the same rank. Higher values indicate better discrimination. All columns are scored on the same union-of-top-genes support ($n_{\text{top}} = 270$).

Cohort	DeSurv (K=3)	Std NMF (K=3)	Matched-budget (K=3, α=0)	Std NMF (K=5, elbow)	Std NMF (K=7, BO α=0)
Dijk	0.607	0.542	0.516	0.633	0.631
Moffitt_GEO_array	0.562	0.521	0.516	0.552	0.578
PACA_AU_array	0.649	0.470	0.462	0.657	0.669
PACA_AU_seq	0.634	0.430	0.412	0.679	0.702
Puleo_array	0.636	0.534	0.533	0.642	0.651

Table S6: Top 270 genes for DeSurv factor D1 (Factor D1 (classical tumor + restCAF-like stroma)), ranked by factor specificity score s_{ij}. The 270-gene cutoff corresponds to the Bayesian optimization–selected projection dimension (n_{top}).

Factor D1 (classical tumor + restCAF-like stroma)					
1–45	46–90	91–135	136–180	181–225	226–270
DMBT1	FGFR2	GOLGA8A	PTPRN2	RPL9	PPP1R12B
IGFBP2	COL9A2	IL2RG	FCGBP	CD79A	PIP5K1B
S100A4	COL22A1	KIAA1211	PLA2G2A	PROM1	AHCYL2
KRT23	IRF7	SFRP1	FABP4	SCD	SST
LIF	SBNO2	ADH1B	EPPK1	TMC5	CYP3A5
IFI6	MGAT3	MFSD2A	CPXM2	CCL14	SMAP2
PDLIM3	SYNM	CLU	KIF1A	PLXNB1	ZG16B
CLDN18	APOD	CLMN	NGFR	TMEM119	SERPINF1
MYO15B	SEMA6A	CHPF	NFKBIA	SORBS2	CLDN3
SYT8	TNXB	MTMR11	SPOCK2	C7	NR1I2
TFF2	PTGIS	TTR	ATP2C2	SLC12A7	JUND
TMPRSS2	CCL18	WFDC2	NBEAL2	QSOX1	NCF2
CHGB	ZC3H12A	C2orf72	CAPN9	UNC5CL	CELSR1
CST1	CELSR3	CCND2	PTGDS	LONRF2	MUC17
B3GNT7	EMB	ARHGAP4	LSP1	VSIG2	INPP5D
F5	KCNJ15	PKDCC	LAMA5	ADAMTSL2	TM4SF5
TNRC18	SORBS1	SRPX	FASN	ARHGEF16	HNF4A
IGF2	PAQR8	CCL21	GFRA1	PCOLCE	CFD
ARSD	SLC4A4	CARD11	PTPRH	CFC1B	HCLS1
ZDHHC7	ACSS1	CTRB2	FMOD	SLC9A3	SH3PXD2B
LENG8	ST6GALNAC1	GOLGA8B	FHL1	ACACB	PCSK2
ATP9A	DES	IYD	PCDH20	BCAS1	HMGCS2
TMPRSS3	PRUNE2	CORO1A	CCR7	RASD1	FILIP1L
CES1	PLCB4	SAA2	CHIT1	LGALS4	ETNK1
HOPX	CADPS	CDCA7	PARM1	MS4A1	TNNI2
CNN1	FMO5	BOC	ITM2A	CAPN6	TOX3
MIA	RAP1GAP2	LTB	NAPSB	SCCPDH	FRZB
CRIP2	GALNT12	COMP	AGT	PTPRCAP	HABP2
NEURL1B	CCL2	MANSC1	FUT3	VNN1	GDA
CHGA	REG3A	SLC19A3	SIPA1L2	PI16	NCOR2
ACTG2	ZCCHC24	STEAP2	SPINK5	SEMA7A	DNAH2
CXCL14	LTF	FOXA1	ACSL1	IL10RA	DOK4
TPPP3	EFHD2	WNK2	BTNL8	GALNT3	HOXB9
PIGR	ATP11A	SELL	CORO2A	GOLM1	KIAA1324
CYP27A1	GP2	PLEKHA6	LAD1	TNFAIP2	TRIM2
ADAM8	HIPK2	NFASC	TMEM63A	CD79B	FGFBP1
FGFR3	SLC26A9	EGR3	GC	ENPP2	IRX3
IGF1R	TACC2	PLIN4	C1orf21	SQLE	TINAGL1
PLEKHB1	TRAK1	WDR72	MIF	KLF4	SFRP4
CPXM1	SAA1	AMY2A	FOXA2	PDLIM7	APLNR
ATP7B	CLDN23	IL16	CCL19	FOSB	AGRN
SLC12A2	FOXJ1	TJP3	GSTA1	RHPN2	SLC25A25
CCL20	FRY	SLC9A2	LLGL2	SCG2	PODN
AKNA	GATA6	LMOD1	ACP5	EMILIN1	ITIH2
COL16A1	PLXNA2	CPS1	UGT2B15	BACE2	FLJ23867

Table S7: Top 270 genes for DeSurv factor D2 (Factor D2 (proCAF stroma)), ranked by factor specificity score s_{ij}.

Factor D2 (proCAF stroma)					
1–45	46–90	91–135	136–180	181–225	226–270
UHMK1	RPS26	SAMD9L	ATP8A1	IRF6	RASSF2
DDR2	MYLK	DOCK2	TGFBR1	KLHL6	ZNF83
INHBA	COL11A1	IGF1	SYNPO2	C1QTNF3	CYTIP
DYNC2H1	NFIB	TACC1	PDE4B	PCLO	SRPX2
ZDHHC20	PTPN13	CFH	SGCD	ITGA2	LY75
ITPR2	PDE3A	DMD	STK17B	LRIG3	HLA-DOA
HMCN1	TRIM22	PCDH7	NUAK1	SSPN	STK38L
LAMA2	DPYD	MALAT1	FCGR3A	F2R	RSAD2
SLFN5	COL14A1	NOTCH2	GPNMB	PDGFC	HBB
PLXNC1	CNTN1	CDK6	VCAM1	CD109	HERC2P2
SVEP1	NCKAP1L	FBN1	ADAMTS9	CLIC4	AOX1
CYBB	PHLDB2	SYNE2	ITGB8	SORL1	PLIN2
SYNE1	ITGA4	RHOBTB3	CR1	AOAH	PFKFB2
SAMHD1	MACC1	CALCRL	SEPT11	FMO2	OGN
SKIL	SEMA5A	NRP1	IFI44L	RDX	DOCK10
ADAMTS12	MRC1	LIFR	ELL2	IL1RAP	RGS4
BICC1	SLIT2	STAB1	GPR183	GBP1	AXL
ITGA1	ZEB2	FLRT2	ZBTB16	CXCR4	FGD6
COL8A1	ADAM10	RNF144A	FRMD6	WNK1	CDC42BPA
PDGFRA	SAMD9	BCAT1	PDE5A	MFAP5	SLC6A6
ABCA1	MS4A7	CILP	CSF1R	ALDH1A3	RGS1
TCF4	EFEMP1	CCDC80	PLEK	CD53	KIAA1324L
CEP170	BIRC3	FAT4	SLK	OMD	CPA3
CD163	WISP1	BNC2	SLC7A2	PLOD2	CHL1
MAN1A1	LAMA4	LTBP1	HEG1	ITGAV	SLC7A11
MSR1	FAM198B	COL12A1	GREM1	MACF1	GFPT2
CCDC88A	PTPRC	IL1R1	VGLL3	NID2	RAI14
F13A1	FAP	SLA	GBP5	NAMPT	LMAN1
RPS28	ARRDC3	MBOAT2	SLC1A3	ADAM28	MECOM
RALGAPA2	MYH10	FNDC3B	FGF7	GNE	FPR1
ZEB1	FCGR2A	PRRX1	ABCA8	ENAH	EDEM3
FBXO32	PARP14	COL10A1	ADAM12	FAM129A	RASSF6
INPP4B	DSE	DDX60	CYBRD1	ECM2	IQGAP2
CDH11	EHF	DOCK9	TFRC	VSIG4	MIAT
PKHD1	OSMR	SRGN	CTSK	MTAP	PRICKLE1
AKAP2	ARID5B	FOXO1	COLEC12	INMT	CHST15
CYP1B1	ASPN	MS4A6A	ALDH1L2	SGMS2	GEM
CCL5	SLCO2B1	ADAMTS2	PGM2L1	COL4A4	CHRDL1
FGL2	FPR3	CPVL	FAM135A	CAPRIN2	DSC2
MAP1B	DOCK8	CD93	DOCK11	CENPF	RAB31
PLXDC2	SLFN11	DST	EPHA3	DCLK1	SLC38A1
LRRK2	ZNF532	CDK14	PAPPA	EDNRA	CNN3
KIAA0754	SPON1	MSRB3	EGFR	WNT5A	FLT1
EDIL3	SEMA3C	RNF213	RASA1	GALNT5	TMEM200A
ITGBL1	FERMT2	SEL1L	AHR	SLC38A2	SLC2A3

Table S8: Top 270 genes for DeSurv factor D3 (Factor D3 (basal-like tumor)), ranked by factor specificity score s_{ij}.

Factor D3 (basal-like tumor)					
1–45	46–90	91–135	136–180	181–225	226–270
HSPB1	HLA-H	LAMB3	MYL9	IL1RN	SDC1
KRT7	HSPA1A	FSCN1	COL17A1	TST	KLK11
MSLN	LGALS3	TSPAN13	EPS8L1	RAPGEFL1	PERP
RPSAP58	FAM83H	TSPAN8	CDH3	C15orf48	ARHGAP23
S100P	EPB41L1	ALDH3B1	RNASET2	PFKP	HPGD
RPS16	EFNA1	ANPEP	TUBA1C	STARD10	AKR1C3
IL32	TKT	GSTP1	RAP1GAP	FOXQ1	CRABP2
SH3BGRL3	SLC2A1	KLK10	H19	REG4	GRB7
IFITM2	PLOD1	B4GALT1	LTBP3	GJB1	SERPINA4
RPS21	IER3	GAS6	AQP3	SLC29A1	SLC39A4
SEMA4B	B3GNT3	ABCC3	MPZL2	NOTCH3	FA2H
APOL1	PRSS3	CLDN2	ID1	TRIM16	RPL8
S100A16	TOB1	CEACAM6	RARRES2	TAP2	MT1X
DNAJB1	EGR1	CREB3L1	SLC25A6	KRT19	TCN1
MT2A	PLA2G16	MST1R	RBP1	CRP	AKAP1
TNFRSF21	OCIAD2	CEBPB	HSPA5	SNHG5	GJB3
RPL28	LPCAT4	TXN	PER1	KRT18	DUOXA2
FXYD3	BST2	NR4A1	ADAP1	CA12	HLA-DRB5
SERPING1	EPHA2	AP1M2	TMC4	MYEOV	ARRDC2
HMGA1	MUC5B	SYNPO	HSD17B2	P4HB	OAS1
S100A14	KCNN4	ANO1	PTP4A1	VILL	ADRA2A
H1F0	ITGA3	SIK1	PLAUR	HKDC1	NQO1
KRT17	TSTA3	EFNB2	ANXA10	STEAP3	LRG1
CSTB	IFI27	CDKN1A	PYGB	PLEK2	SERPINH1
ASS1	MMP1	S100A10	CYP2S1	CD68	COL18A1
EFNB1	SLPI	PKP3	ALDOA	MAFF	CERCAM
MDK	CLTB	MCM7	CDCP1	CD9	CA9
LY6E	SPINK1	RARRES3	IFITM3	PLLP	CDHR2
CAMK2N1	JUP	CEACAM1	BAIAP2L1	EPHB4	ESRP1
MGLL	PPP1R15A	ALDH2	MGST1	KRT16	CRYZ
PDZK1IP1	TRIM29	CES2	TFF1	ABLIM3	TAGLN2
BCL2L1	PPP1R1B	SLC6A8	MUC4	PTPRU	HNF1B
C19orf33	ARPC1A	PPL	MMP28	MUC6	BMP4
SFN	GPRC5A	NDRG1	CLDN1	FDFT1	MT1E
ITGB4	RAB25	AZGP1	TM4SF1	GAPDH	ANGPTL4
TSPAN1	ZFP36L2	S100A11	TFF3	EPS8L3	CFB
SLC16A3	CLDN7	SEMA3B	UNC13D	CLDN10	GDF15
C4A	DDIT4	TNFRSF12A	SERINC2	ITPKC	PADI1
KLF5	FKBP4	SOX9	SPNS2	LOXL2	TMSB10
IER2	LAPTM4B	PMEPA1	AQP5	AMIGO2	B2M
ZFP36L1	MUC13	CYR61	IGFBP4	DUOX2	CAV2
MAL2	PSCA	EPCAM	SLC7A5	CLDN4	CTSH
SCNN1A	SDCBP2	BHLHE40	TESC	CAPG	PABPC1
MALL	PROM2	CX3CL1	IFITM1	CEACAM5	KRT8
BGN	LRRC8A	SMAD3	KCNK5	VWA1	TAP1